

Deep Frontier AI Analysis

Reference date: 2026-05-15. Generated at: 2026-07-12T19:42:51+00:00.

This report is deliberately data-heavy. It uses the local rich frontier-model dataset, the public Anthropic Economic Index release files for occupation exposure, and a new domain benchmark layer covering coding, medicine, terminal agents, finance, legal reasoning, math, science/reasoning, language, vision and search/document work. The goal is not to claim precision about the future; it is to make the assumptions inspectable enough that the forecast can be argued with.

Dashboard Snapshot

This opening map is the fast path through the analysis. It turns the long report into a set of inspectable questions, each tied to a primary artifact and an evidence-strength label.

section	question	headline	metric	evidence_level	primary_artifact
Models	Who leads the current frontier-family signal?	GPT	78.2 heuristic index	observed	company_frontier_scores.csv
Domains	Which capability field is improving fastest in the public panel?	Agentic terminal work	12.0 points/year used	scenario	domain_improvement_velocity.csv
Coverage	How many capability domains have broad benchmark coverage?	1 broad domains	115,938 normalized rows	observed	domain_benchmark_catalog.csv
Evidence	Where is model-level evidence strongest?	GPT	120 direct benchmark matches	direct_match	family_coverage_matrix.csv
Forecast	Who wins the 10-year frontier-quality stress test?	GPT	58.6% simulation share	scenario	company_next_frontier_probabilities.csv
Open Ecosystem	Who benefits if distribution, openness and cost matter more?	Qwen	70.2% simulation share	scenario	company_next_frontier_probabilities.csv
Domain Forecast	Which field has the highest 2036 base-case frontier score?	Software engineering	97.4/100 forecast score	scenario	domain_capability_forecasts.csv
Open vs Closed	Where is the open-vs-closed gap largest?	text to image	361.6 arena rating points	observed	open_closed_gap_by_category.csv
Economics	Which directly matched model sits highest on price-performance?	OpenAI: GPT-4.1 Nano	92.2 direct index	direct_match	direct_model_price_performance.csv
Economics	Is average message/task cost rising in the modeled workload mix?	0.152 USD	577.1 index vs 2023	scenario	llm_message_cost_trends.csv
Fixed Task Cost	What happens to a thesis-quality fixed writing task?	GPT	0.0002 USD in 2036	scenario	fixed_task_cost_curves.csv
Labor	Which occupation has the highest near-term pressure index?	Data Entry Keyers	60.2 disruption index	observed	job_exposure_scores.csv
Labor	Where is whole-job replacement most feasible after gates?	Market Research Analysts and Marketing Specialists	39.0 feasibility index	observed	job_replacement_feasibility.csv
Workflows	Which business domain should a reader inspect first?	finance analysis	38.1 disruption index	family_proxy	business_domain_ai_pressure.csv
Execution	Which family shows the most visible recent release velocity?	GPT	66 releases in 365 days	observed	release_cadence_by_family.csv
Coverage	How fresh is the visible source layer?	2026-05-15	1,327,480 source rows tracked	observed	source_coverage_diagnostics.csv
Uncertainty	How many family ranks are stable under evidence-scaled stress?	5 stable rank bands	11 families stress-tested	scenario	rank_stability_intervals.csv
Risk	What should a reviewer challenge first?	3 high-severity assumptions	Named failure modes	speculative	claim_failure_modes.csv

How To Read This Report

The report is organized around three questions:

- 1 Who has the strongest frontier-family signal right now? The answer is a composite heuristic, so the report shows both rank and component composition instead of hiding the weighting.
- 1 Where are the counterintuitive gaps? Open-weight systems, low prices, context windows and benchmark ratings move on different axes. The plots keep those axes separate.
- 1 Which domains are improving fastest? The domain panel keeps fields separate: coding and agentic terminal work should not be averaged blindly with medicine, legal reasoning or finance.
- 1 What happens when model capability meets labor structure? Occupation exposure is not the same thing as replacement. The labor section separates task pressure, augmentation, bottlenecks and whole-job feasibility.
- 1 Are costs rising or falling? The economics section separates workload-mix cost per message/task from fixed-task cost curves, because those can move in opposite directions.

Every chart should be read as an audit surface. If a conclusion depends on one metric, the report names that metric and shows the caveat near the visualization. Domain rows with `forecast_enabled=false` are deliberately held flat: no comparable history means no extrapolation. The full remediation log is in `docs/statistical_audit.md`.

Evidence badges used throughout the HTML view: `observed`, `direct_match`, `family_proxy`, `scenario`, `speculative`. They are labels for evidence strength, not decoration.

Executive Takeaways

- 1 Near-term frontier-family leadership is concentrated, but not one-dimensional. The highest heuristic index in this run is GPT with a frontier momentum heuristic index of 78.2. The strongest openness/cost/ecosystem signal is Qwen, which is not automatically the same thing as best closed frontier performance.
- 1 The next-winner question is a simulation sensitivity exercise. The table changes component weights thousands of times and injects evidence noise. Its shares are not calibrated probabilities.
- 1 Open vs closed is category-specific. Some LMArena categories show narrow gaps; others preserve a clear closed/API advantage. "Open source caught up" is too crude.
- 1 Field-level progress is uneven. Coding, terminal-agent and language/document signals have denser coverage than legal and finance. The report extrapolates only domains with repeated observations of the same benchmark; other domains are marked `insufficient_history` and held flat.
- 1 The job story is not "all jobs disappear." The highest-risk roles are task bundles where language, analysis, clerical transformation and directive delegation are already exposed. Jobs with physical work, trust, regulation or face-to-face accountability keep meaningful bottlenecks.
- 1 The 10-year labor path is a scenario, not an estimate. The task-contact paths encode explicit adoption assumptions; they are useful for stress testing verification, liability and workflow redesign, not for predicting employment levels.
- 1 The cost view is synthetic. Message/task paths combine assumed workload mixes with current catalog cohorts, while fixed-task paths use explicit quality and price scenarios. Neither is observed invoice history.

Data Freshness And Coverage

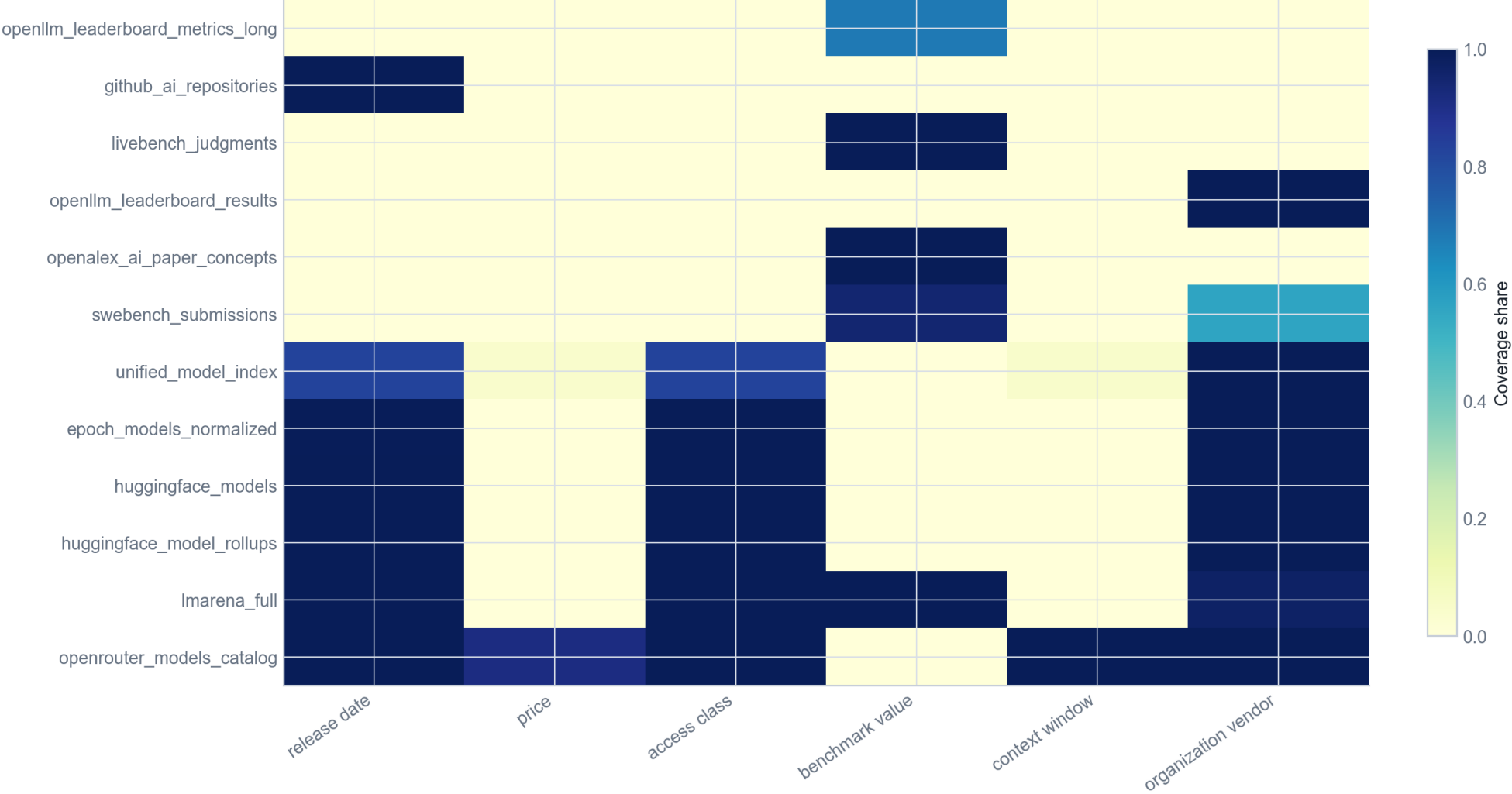
The report now exposes source coverage before leaning on rankings. This section records row counts, captured dates, latest source dates and missingness across release dates, prices, benchmark values, context windows and organization/vendor fields.

table	rows	latest_source_date	release_date_coverage	price_coverage	benchmark_value_coverage	organization_vendor_coverage
epoch_models_normalized	3525	2026-04-24	0.9935	0	0	1
github_ai_repositories	1395	2026-05-15	1	0	0	0
github_ai_repository_topics	13746		0	0	0	0
github_model_mentions	703		0	0	0	0
huggingface_model_files	5256		0	0	0	0
huggingface_model_rollups	2574	2026-05-15	1	0	0	1
huggingface_model_tags	36737		0	0	0	0
huggingface_models	2574	2026-05-15	1	0	0	1
livebench_judgments	60372		0	0	1	0
lmarena_full	862027	2026-05-14	1	0	1	0.9726
openalex_ai_paper_authorships	22643		0	0	0	0
openalex_ai_paper_concepts	36079		0	0	1	0

Family coverage matrix:

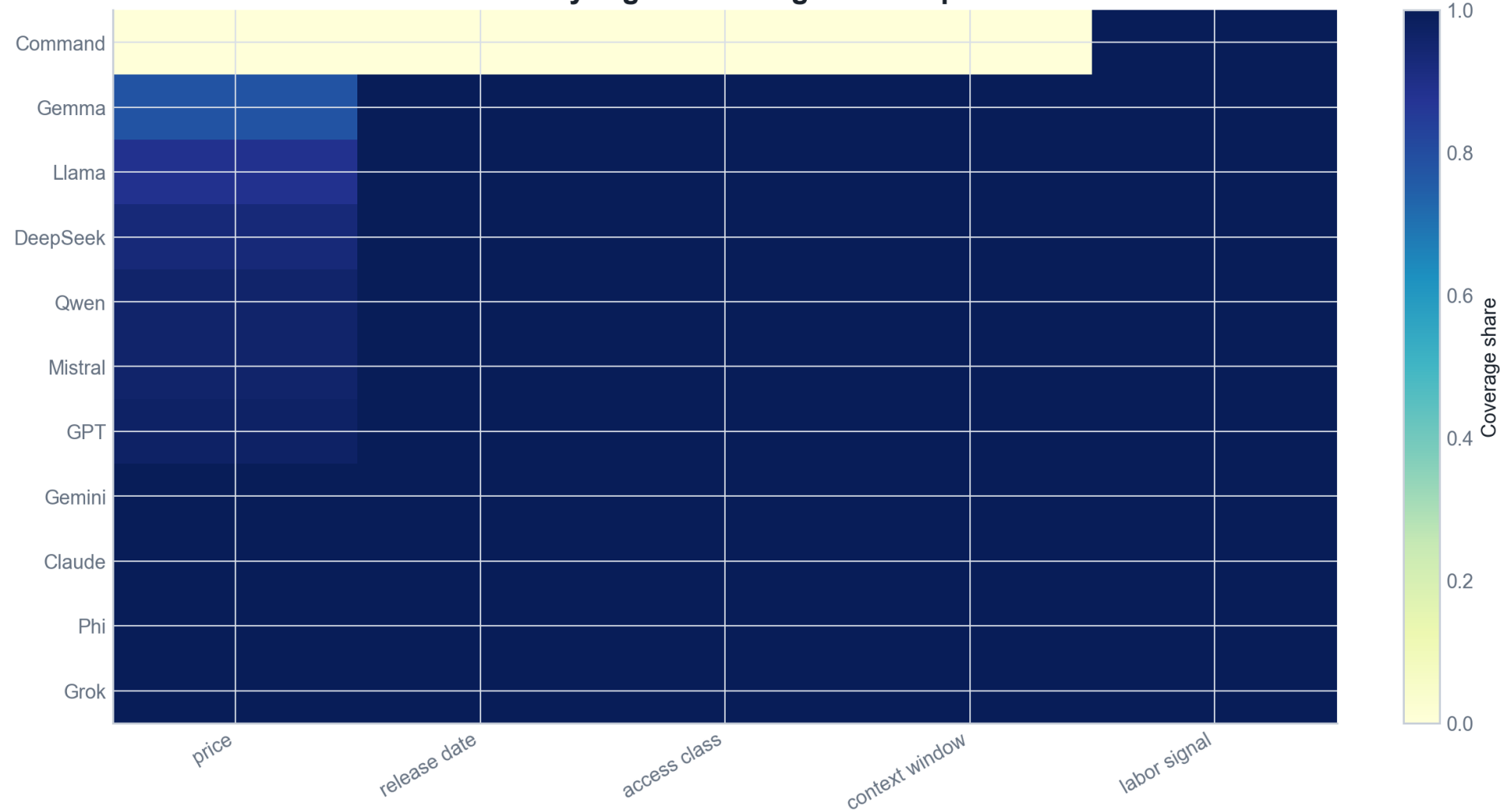
model_family	vendor	coverage_score	direct_benchmark_match_count	family_proxy_benchmark_count	source_gap_count
Gemini	Google	100	24	32	0
Claude	Anthropic	100	48	30	0
Phi	Microsoft	100	6	8	0
Grok	xAI	100	3	24	0
GPT	OpenAI	99.5	120	61	0
Mistral	Mistral	99.33	56	23	0
Qwen	Alibaba	99.3	83	52	0
DeepSeek	DeepSeek	98.89	25	20	0
Llama	Meta	98.08	81	25	0
Gemma	Google	96.3	15	14	0
Command	Command	0	0	0	6

Source Freshness and Coverage Dashboard



Source coverage dashboard

Family Signal Coverage Heatmap



Family signal coverage heatmap

Capability Domains

This is the new domain benchmark layer. It pulls together local benchmark sources and additional public sources downloaded during generation: LiveCodeBench, Open Medical-LLM Leaderboard result files, Terminal-Bench, FinanceBench, QFBench and Lexometrica LegalBench RU. Scores are normalized to a 0-100 frontier scale so fields can be compared without pretending that a medical QA percent, a legal composite, an arena rating and an agentic terminal score are the same measurement.

Domain catalog:

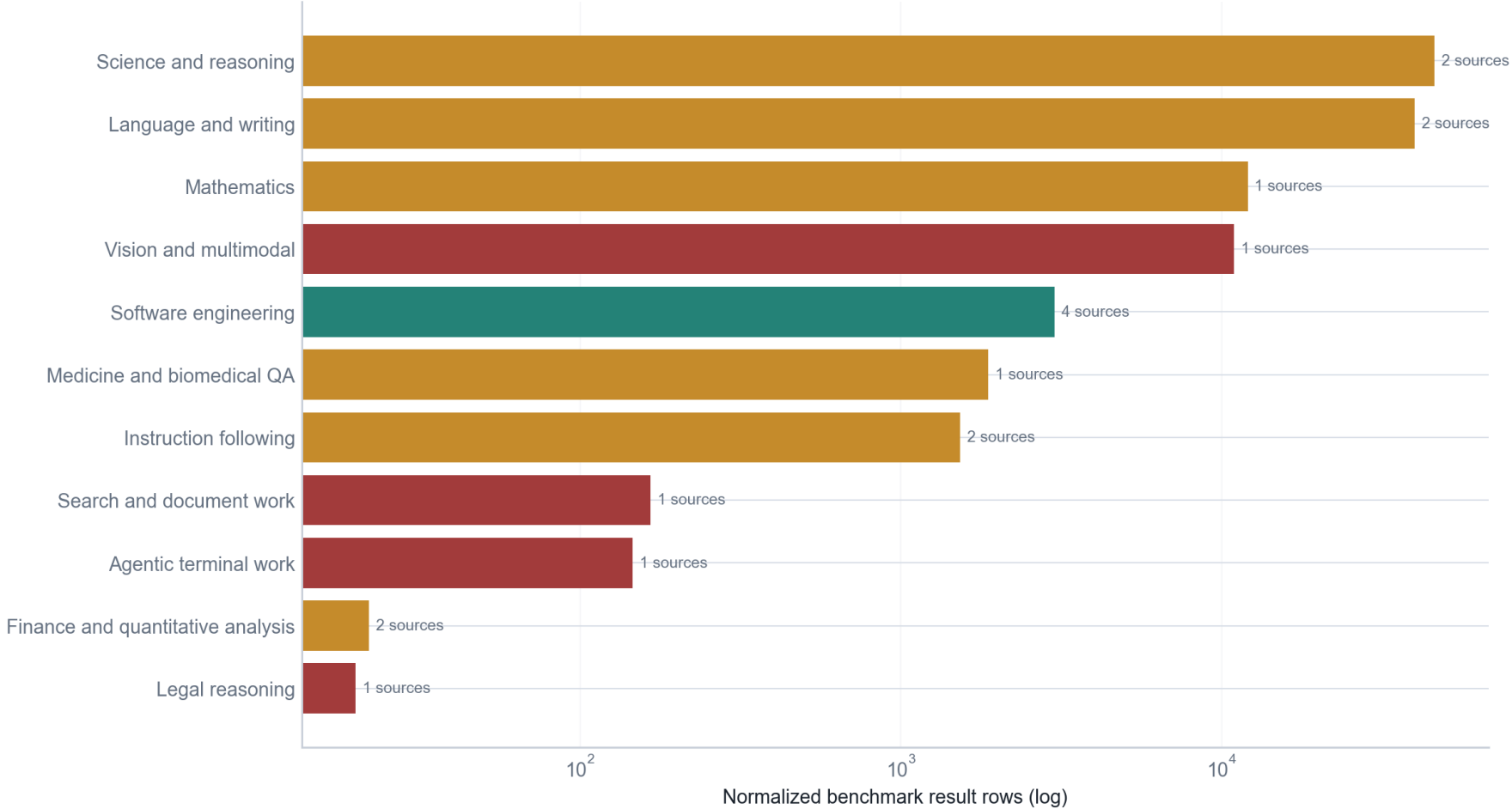
domain_label	normalized_result_rows	source_count	benchmark_count	model_count	coverage_label	latest_eval_date	interpretation
Software engineering	3020	4	4	391	broad	2026-05-14	Code generation, repository repair, web development and terminal software workflows.
Science and reasoning	46146	2	36	1368	moderate	2026-05-12	Graduate-level science, GPQA-like reasoning, ARC/BBH/MuSR and broad reasoning suites.
Language and writing	40009	2	2	480	moderate	2026-05-14	General text quality, paraphrase, editing, summarization and subjective chat preference.
Instruction following	1529	2	2	1457	moderate	2025-04-07	Constraint following, output format obedience and prompt-level generalization.
Finance and quantitative analysis	22	2	3	13	moderate	2026-05-07	Financial QA, quantitative coding, risk, pricing, forecasting and professional finance tasks.
Mathematics	12060	1	9	1339	moderate		Competition math, quantitative reasoning and formal problem solving.
Medicine and biomedical QA	1873	1	10	184	moderate	2025-01-29	Medical question answering, biomedical literature reasoning and clinical knowledge subsets.
Vision and multimodal	10947	1	3	202	thin	2026-05-12	Image understanding, image editing, text-to-image and multimodal preference leaderboards.
Search and document work	166	1	1	24	thin	2026-05-12	Search, long-document handling, retrieval-facing work and document synthesis.
Agentic terminal work	146	1	1	51	thin	2026-05-15	Long-horizon command-line tasks requiring planning, execution, debugging and environment control.
Legal reasoning	20	1	2	10	thin	2026-03-01	Legal issue spotting, rule application, citations and jurisdiction-specific legal reasoning.

Representative high-scoring source rows:

source_name	domain_label	benchmark	task	model_name	score_normalized_0_100	eval_date	limitations
Terminal-Bench 2.0 leaderboard	Agentic terminal work	Terminal-Bench 2.0	vix	Claude Opus 4.7	90.2	2026-05-15	Agent, scaffold and model are entangled; do not attribute the whole score to model weights alone.
Terminal-Bench 2.0 leaderboard	Agentic terminal work	Terminal-Bench 2.0	JJAgent	Multiple	87.1	2026-05-15	Agent, scaffold and model are entangled; do not attribute the whole score to model weights alone.
Terminal-Bench 2.0 leaderboard	Agentic terminal work	Terminal-Bench 2.0	NexAU-AHE	GPT-5.5	84.7	2026-05-14	Agent, scaffold and model are entangled; do not attribute the whole score to model weights alone.
Terminal-Bench 2.0 leaderboard	Agentic terminal work	Terminal-Bench 2.0	LemonHarness	Multiple	84.5	2026-05-14	Agent, scaffold and model are entangled; do not attribute the whole score to model weights alone.
Terminal-Bench 2.0 leaderboard	Agentic terminal work	Terminal-Bench 2.0	Capy	GPT-5.5	83.1	2026-05-14	Agent, scaffold and model are entangled; do not attribute the whole score to model weights alone.
Terminal-Bench 2.0 leaderboard	Agentic terminal work	Terminal-Bench 2.0	Polaris	Multiple	82.2	2026-05-14	Agent, scaffold and model are entangled; do not attribute the whole score to model weights alone.

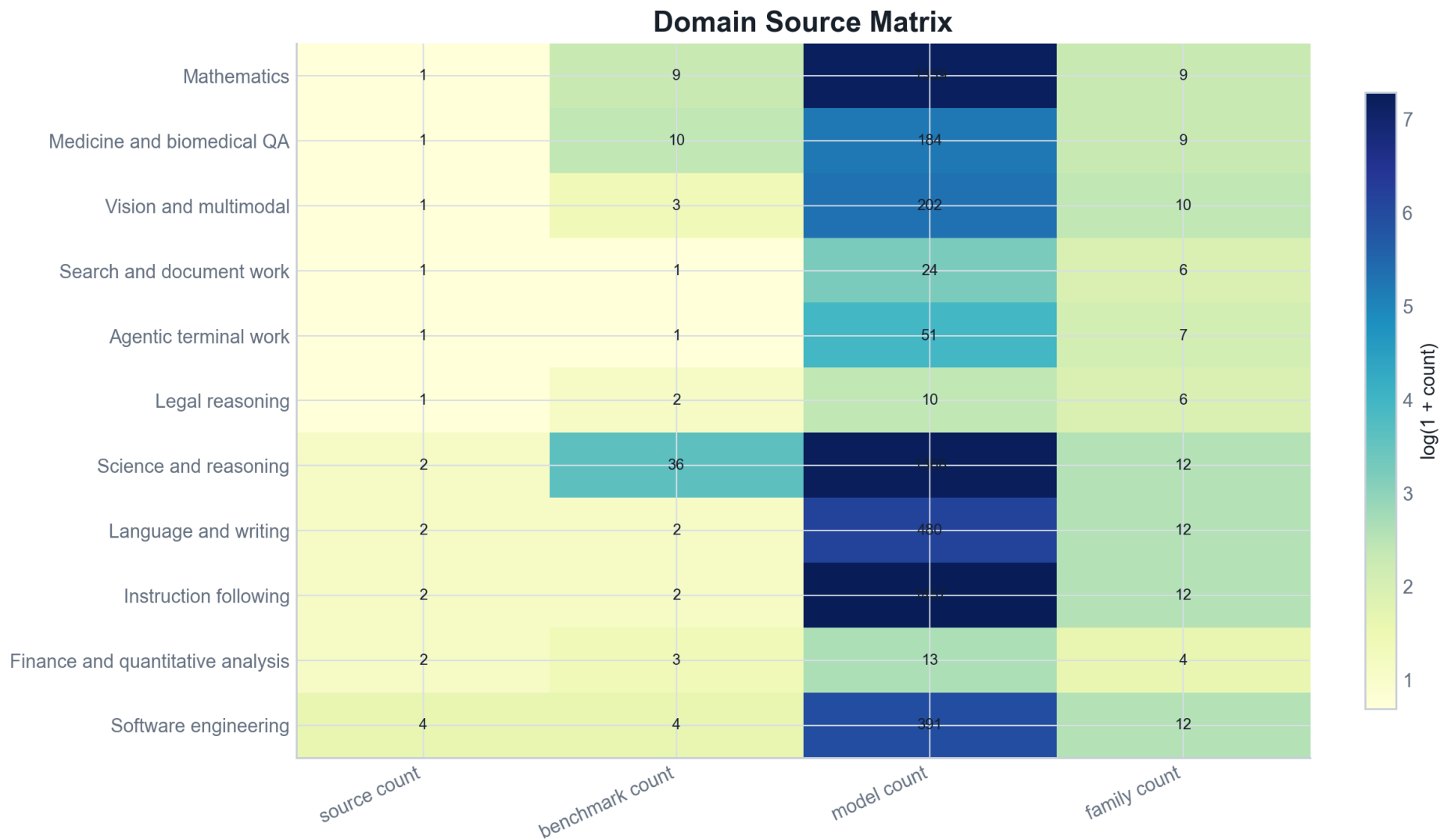
source_name	domain_label	benchmark	task	model_name	score_normalized_0_100	eval_date	limitations
Terminal-Bench 2.0 leaderboard	Agentic terminal work	Terminal-Bench 2.0	Codex CLI	GPT-5.5	82	2026-04-23	Agent, scaffold and model are entangled; do not attribute the whole score to model weights alone.
Terminal-Bench 2.0 leaderboard	Agentic terminal work	Terminal-Bench 2.0	ForgeCode	GPT-5.4	81.8	2026-03-12	Agent, scaffold and model are entangled; do not attribute the whole score to model weights alone.
Terminal-Bench 2.0 leaderboard	Agentic terminal work	Terminal-Bench 2.0	WOZCODE	Claude Opus 4.7	80.2	2026-05-14	Agent, scaffold and model are entangled; do not attribute the whole score to model weights alone.
Terminal-Bench 2.0 leaderboard	Agentic terminal work	Terminal-Bench 2.0	TongAgents	Gemini 3.1 Pro	80.2	2026-03-13	Agent, scaffold and model are entangled; do not attribute the whole score to model weights alone.
Terminal-Bench 2.0 leaderboard	Agentic terminal work	Terminal-Bench 2.0	LemonHarness	Multiple	79.9	2026-05-14	Agent, scaffold and model are entangled; do not attribute the whole score to model weights alone.
Terminal-Bench 2.0 leaderboard	Agentic terminal work	Terminal-Bench 2.0	ForgeCode	Claude Opus 4.6	79.8	2026-03-12	Agent, scaffold and model are entangled; do not attribute the whole score to model weights alone.
Terminal-Bench 2.0 leaderboard	Agentic terminal work	Terminal-Bench 2.0	SageAgent	GPT-5.3-Codex	78.4	2026-03-13	Agent, scaffold and model are entangled; do not attribute the whole score to model weights alone.
Terminal-Bench 2.0 leaderboard	Agentic terminal work	Terminal-Bench 2.0	ForgeCode	Gemini 3.1 Pro	78.4	2026-03-02	Agent, scaffold and model are entangled; do not attribute the whole score to model weights alone.
Terminal-Bench 2.0 leaderboard	Agentic terminal work	Terminal-Bench 2.0	Droid	GPT-5.3-Codex	77.3	2026-02-24	Agent, scaffold and model are entangled; do not attribute the whole score to model weights alone.
Terminal-Bench 2.0 leaderboard	Agentic terminal work	Terminal-Bench 2.0	Meta-Harness	Claude Opus 4.6	76.4	2026-05-14	Agent, scaffold and model are entangled; do not attribute the whole score to model weights alone.
Terminal-Bench 2.0 leaderboard	Agentic terminal work	Terminal-Bench 2.0	CodeBrain-1.5	GPT-5.3-Codex	75.8	2026-02-10	Agent, scaffold and model are entangled; do not attribute the whole score to model weights alone.
Terminal-Bench 2.0 leaderboard	Agentic terminal work	Terminal-Bench 2.0	Codelia	GPT-5.3-Codex	75.7	2026-05-14	Agent, scaffold and model are entangled; do not attribute the whole score to model weights alone.

Benchmark Coverage by Capability Domain



Coverage is deliberately visible because domain forecasts are only as useful as the source breadth behind them.

Domain benchmark coverage



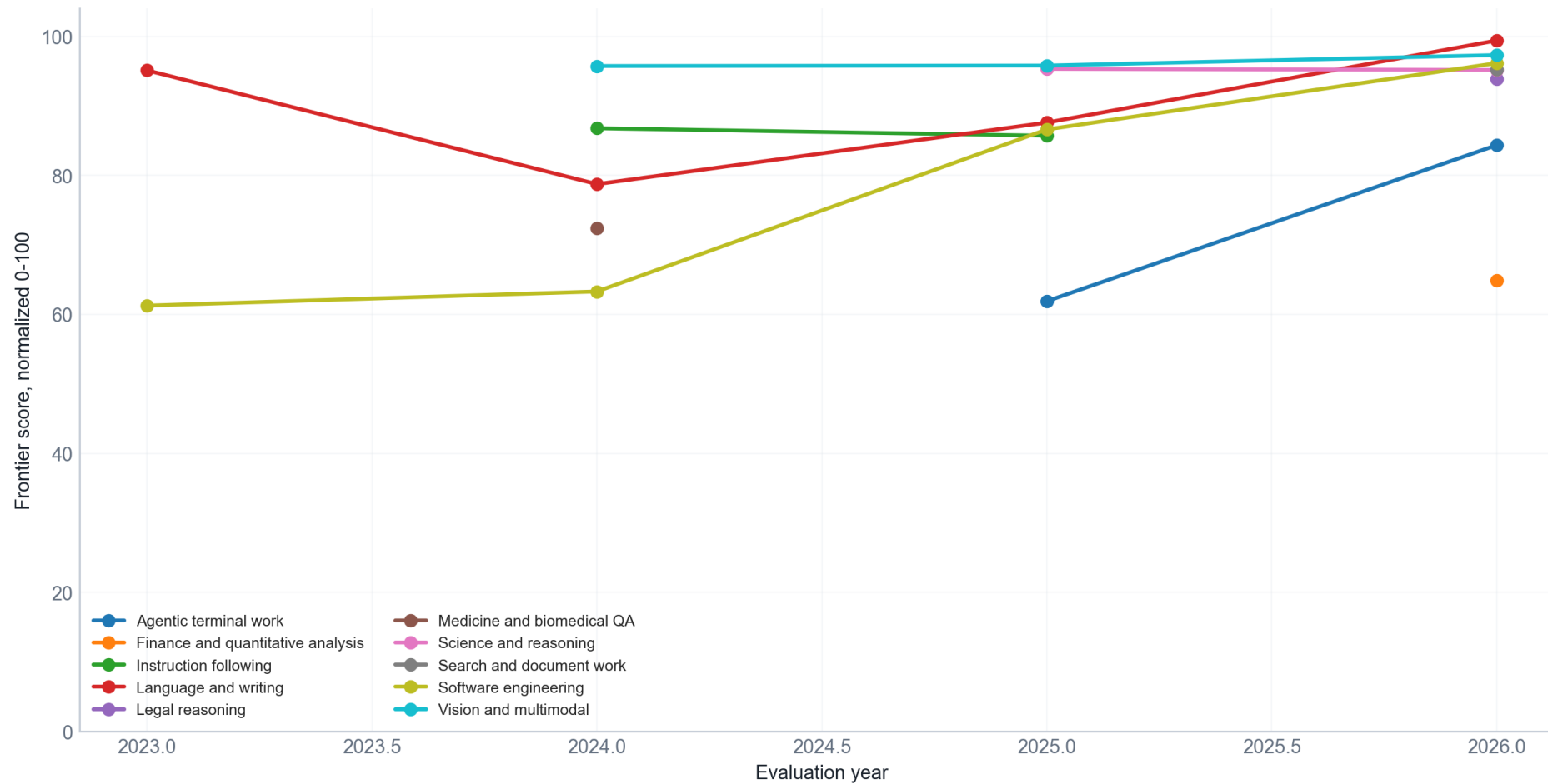
Domain source matrix

Domain Improvement Velocity

The velocity table estimates how quickly each field is improving in the public benchmark panel. When a domain has enough dated observations, the report uses its observed frontier slope. When the time series is too thin, it falls back to the cross-domain median and labels the slope source explicitly.

domain_label	current_frontier_score	annual_frontier_point_gain_used	slope_source	years_observed	coverage_label	forecast_confidence
Vision and multimodal	97.31	1.542	median_within_benchmark_slope	3	thin	low
Search and document work	95.21	0	insufficient_comparable_history	1	thin	insufficient_history
Science and reasoning	95.17	0	median_within_benchmark_slope	2	moderate	insufficient_history
Legal reasoning	93.88	0	insufficient_comparable_history	1	thin	insufficient_history
Language and writing	87.71	9.573	median_within_benchmark_slope	4	moderate	low
Instruction following	85.71	0	median_within_benchmark_slope	2	moderate	insufficient_history
Agentic terminal work	84.36	12	median_within_benchmark_slope	2	thin	low
Software engineering	79.87	4.094	median_within_benchmark_slope	4	broad	medium
Medicine and biomedical QA	72.39	0	insufficient_comparable_history	1	moderate	insufficient_history
Finance and quantitative analysis	64.89	0	insufficient_comparable_history	1	moderate	insufficient_history
Mathematics	39.38	0	insufficient_comparable_history	0	moderate	insufficient_history

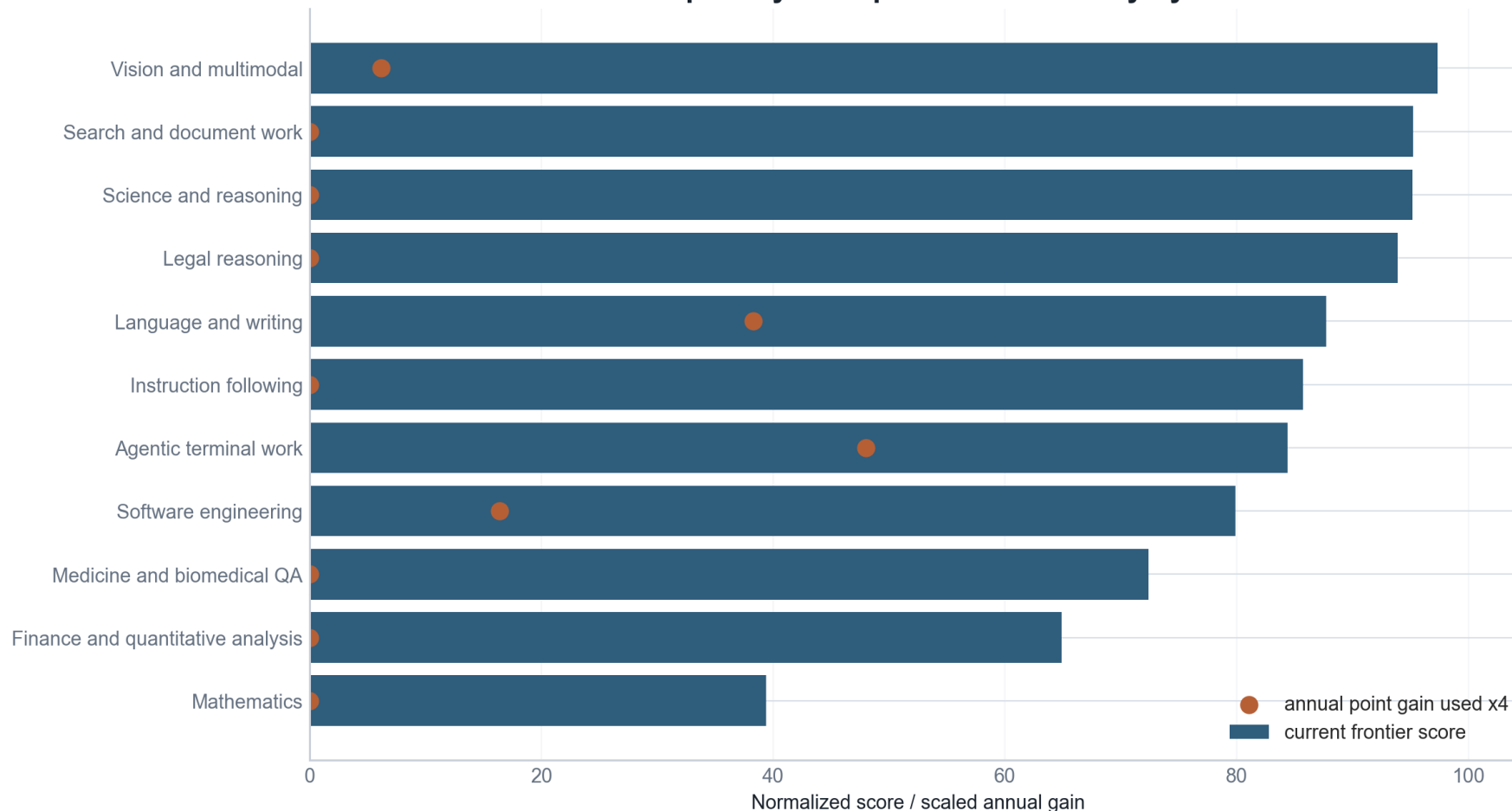
Observed Domain Frontier Trend



Scores are normalized across heterogeneous benchmarks. The trend is useful for direction and relative velocity, not exact cross-domain psychometrics.

Domain frontier trends

Current Capability vs Improvement Velocity by Domain



Orange dots are scaled so velocity can be read beside current level. Thin domains use the cross-domain fallback and are labeled in the table.

Domain current velocity

Domain Capability Forecasts

The domain forecast uses a bounded gap-closure model: a domain starts at its current normalized frontier score, closes a fraction of the remaining gap each year, and is capped below 100. This makes the forecast interpretable: the question is how quickly each field closes the remaining gap, not whether scores can grow without limit.

Base scenario by domain and horizon:

domain_label	target_year	forecast_frontier_score	current_frontier_score	confidence	caveat
Vision and multimodal	2028	98.17	97.31	low	The data mixes perception and generation; downstream reliability depends heavily on task framing.
Vision and multimodal	2031	98.97	97.31	low	The data mixes perception and generation; downstream reliability depends heavily on task framing.
Vision and multimodal	2036	99.5	97.31	low	The data mixes perception and generation; downstream reliability depends heavily on task framing.
Search and document work	2028	95.21	95.21	insufficient_history	Retrieval quality, source grounding and tool access can dominate model-only scores.
Search and document work	2031	95.21	95.21	insufficient_history	Retrieval quality, source grounding and tool access can dominate model-only scores.
Search and document work	2036	95.21	95.21	insufficient_history	Retrieval quality, source grounding and tool access can dominate model-only scores.
Science and reasoning	2028	95.17	95.17	insufficient_history	This is a mixed domain; gains may come from either knowledge, search, reasoning-time or benchmark-specific training.
Science and reasoning	2031	95.17	95.17	insufficient_history	This is a mixed domain; gains may come from either knowledge, search, reasoning-time or benchmark-specific training.
Science and reasoning	2036	95.17	95.17	insufficient_history	This is a mixed domain; gains may come from either knowledge, search, reasoning-time or benchmark-specific training.
Legal reasoning	2028	93.88	93.88	insufficient_history	Legal performance is highly jurisdictional; benchmark score is not professional legal authority.
Legal reasoning	2031	93.88	93.88	insufficient_history	Legal performance is highly jurisdictional; benchmark score is not professional legal authority.
Legal reasoning	2036	93.88	93.88	insufficient_history	Legal performance is highly jurisdictional; benchmark score is not professional legal authority.
Language and writing	2028	97.09	87.71	low	Human preference and style vary; benchmark gains do not map one-to-one to brand-safe writing quality.
Language and writing	2031	99.5	87.71	low	Human preference and style vary; benchmark gains do not map one-to-one to brand-safe writing quality.
Language and writing	2036	99.5	87.71	low	Human preference and style vary; benchmark gains do not map one-to-one to brand-safe writing quality.
Instruction following	2028	85.71	85.71	insufficient_history	High scores can hide brittle behavior on a user's own constraints, so local evals remain important.
Instruction following	2031	85.71	85.71	insufficient_history	High scores can hide brittle behavior on a user's own constraints, so local evals remain important.
Instruction following	2036	85.71	85.71	insufficient_history	High scores can hide brittle behavior on a user's own constraints, so local evals remain important.
Agentic terminal work	2028	96.29	84.36	low	Agent scaffolding can dominate raw model quality, so model and agent should be separated when possible.
Agentic terminal work	2031	99.5	84.36	low	Agent scaffolding can dominate raw model quality, so model and agent should be separated when possible.

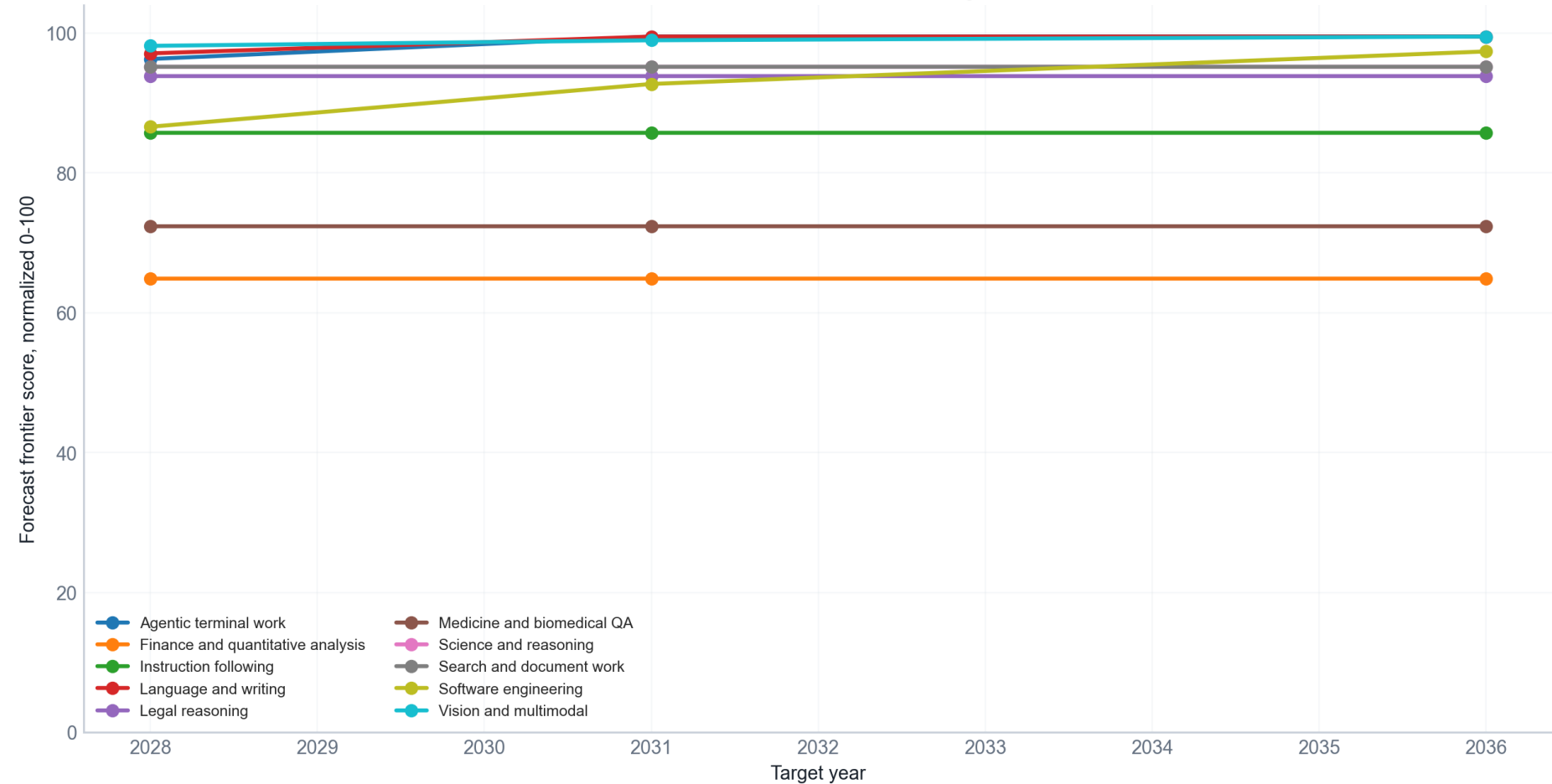
domain_label	target_year	forecast_frontier_score	current_frontier_score	confidence	caveat
Agentic terminal work	2036	99.5	84.36	low	Agent scaffolding can dominate raw model quality, so model and agent should be separated when possible.
Software engineering	2028	86.6	79.87	medium	Coding benchmarks move quickly and are contamination-sensitive; use fresh task windows when possible.
Software engineering	2031	92.72	79.87	medium	Coding benchmarks move quickly and are contamination-sensitive; use fresh task windows when possible.
Software engineering	2036	97.37	79.87	medium	Coding benchmarks move quickly and are contamination-sensitive; use fresh task windows when possible.
Medicine and biomedical QA	2028	72.39	72.39	insufficient_history	Multiple-choice medical QA is not clinical deployment safety; human review and liability remain binding.
Medicine and biomedical QA	2031	72.39	72.39	insufficient_history	Multiple-choice medical QA is not clinical deployment safety; human review and liability remain binding.
Medicine and biomedical QA	2036	72.39	72.39	insufficient_history	Multiple-choice medical QA is not clinical deployment safety; human review and liability remain binding.
Finance and quantitative analysis	2028	64.89	64.89	insufficient_history	Benchmarks are sparse and often workflow-specific; treat forecasts as directional until more longitudinal data exists.
Finance and quantitative analysis	2031	64.89	64.89	insufficient_history	Benchmarks are sparse and often workflow-specific; treat forecasts as directional until more longitudinal data exists.
Finance and quantitative analysis	2036	64.89	64.89	insufficient_history	Benchmarks are sparse and often workflow-specific; treat forecasts as directional until more longitudinal data exists.

Threshold timing:

domain_label	threshold_score	base_years_to_threshold	estimated_threshold_year	confidence
Vision and multimodal	90	0.0	2026.0	low
Vision and multimodal	95	0.0	2026.0	low
Search and document work	90	0.0	2026.0	insufficient_history
Search and document work	95	0.0	2026.0	insufficient_history
Science and reasoning	90	0.0	2026.0	insufficient_history
Science and reasoning	95	0.0	2026.0	insufficient_history
Legal reasoning	90	0.0	2026.0	insufficient_history
Legal reasoning	95	n/a	n/a	insufficient_history
Language and writing	90	0.29	2027.0	low
Language and writing	95	1.25	2028.0	low
Instruction following	90	n/a	n/a	insufficient_history
Instruction following	95	n/a	n/a	insufficient_history
Agentic terminal work	90	0.62	2027.0	low
Agentic terminal work	95	1.58	2028.0	low
Software engineering	90	3.44	2030.0	medium
Software engineering	95	6.85	2033.0	medium
Medicine and biomedical QA	90	n/a	n/a	insufficient_history
Medicine and biomedical QA	95	n/a	n/a	insufficient_history

domain_label	threshold_score	base_years_to_threshold	estimated_threshold_year	confidence
Finance and quantitative analysis	90	n/a	n/a	insufficient_history
Finance and quantitative analysis	95	n/a	n/a	insufficient_history
Mathematics	90	n/a	n/a	insufficient_history
Mathematics	95	n/a	n/a	insufficient_history

Base Scenario Domain Capability Forecast



Forecasts use bounded gap closure from observed domain trends. They become more speculative when source coverage is thin.

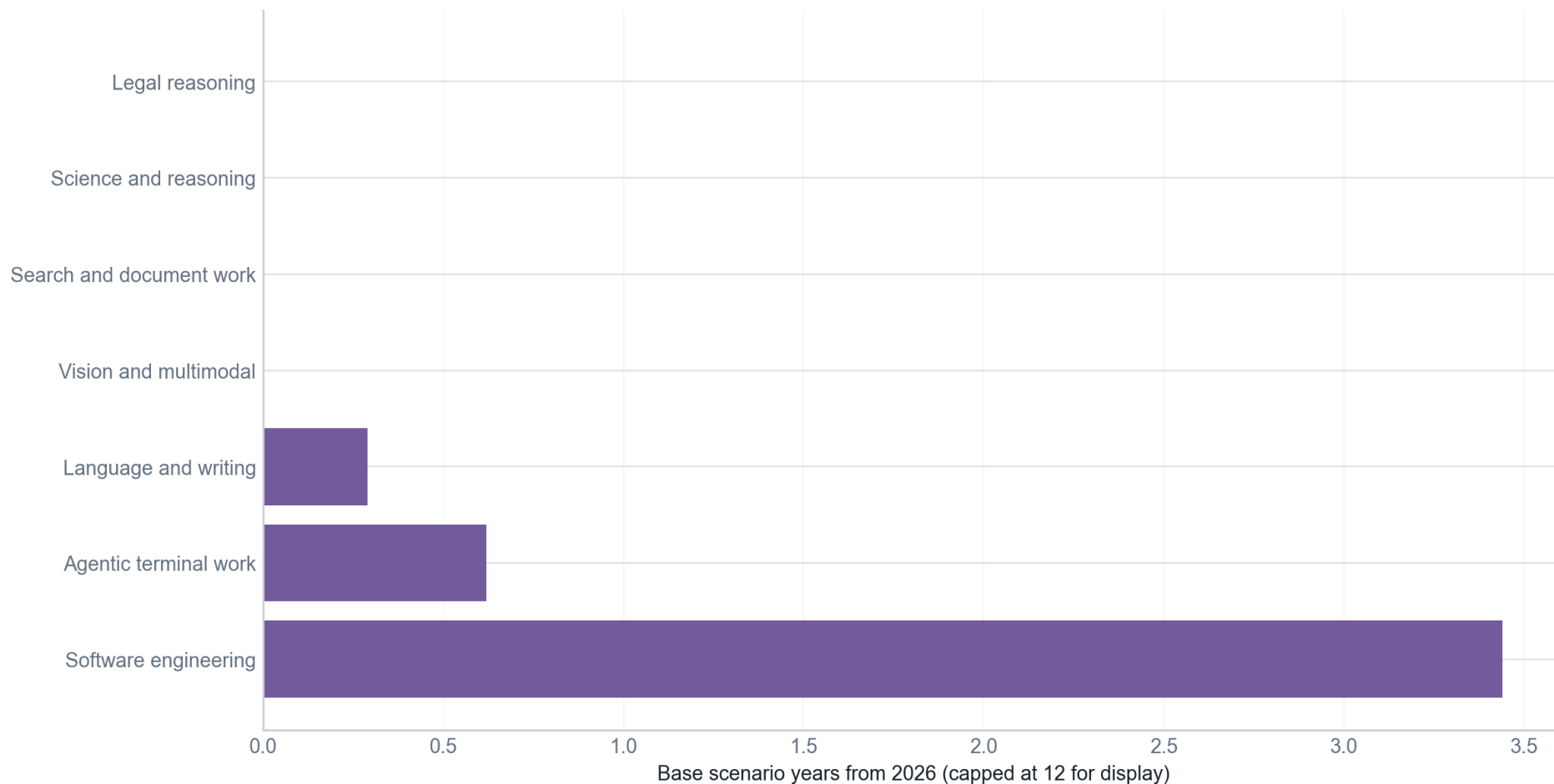
Domain forecast base

Domain Forecast Scenario Small Multiples



Domain forecast scenarios

Estimated Years to 90/100 Domain Frontier Score



A zero means the normalized 90 threshold is already observed in this public benchmark panel. Thin-domain estimates should be read cautiously.

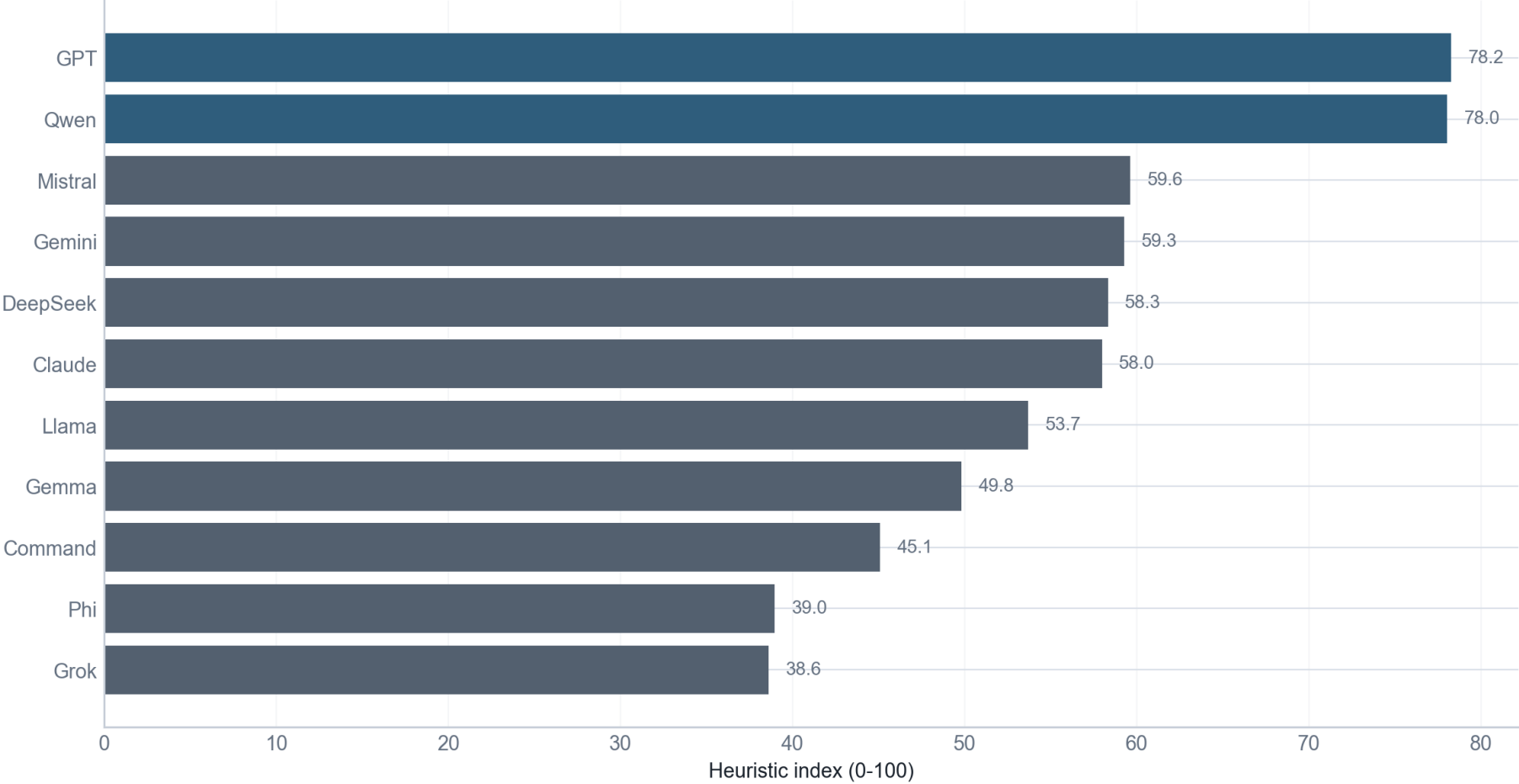
Domain threshold timeline

Model Family Frontier Score

The index ranks model families and product lines, not legal companies. It blends benchmark performance, release velocity, API surface, price, research/ecosystem pull and openness. It is not a universal truth; sensitivity outputs show which rankings are weight-sensitive.

rank	model_family	frontier_momentum_heuristic_index	sensitivity_label	performance_component	release_velocity_component	ecosystem_component	cost_efficiency_component	openness_component
1	GPT	78.25	stable_top_tier	69.2487	100	87.1657	91.7046	41.4415
2	Qwen	78.05	stable_top_tier	74.6117	86.7888	80.9259	94.6247	97.9167
3	Mistral	59.62	weight_sensitive	60.7925	35.7759	70.6138	100	95.1064
4	Gemini	59.26	weight_sensitive	77.0602	49.903	60.3614	80.9676	0
5	DeepSeek	58.32	weight_sensitive	48.7632	32.9634	78.004	85.8923	92.2704
6	Claude	57.97	weight_sensitive	82.4818	35.8405	57.7862	36.1209	55
7	Llama	53.67	weight_sensitive	36.747	2.8125	81.3439	100	91.607
8	Gemma	49.79	weight_sensitive	53.4005	8.50216	65.6807	96.1247	91.9149
9	Command	45.08	weight_sensitive	50	50	25	50	41.6667
10	Phi	38.95	weight_sensitive	37.6984	0	48.871	91.7046	83.1769
11	Grok	38.62	weight_sensitive	65.0893	23.4806	27.389	0	1.84211

Frontier Momentum Heuristic Index by Model Family

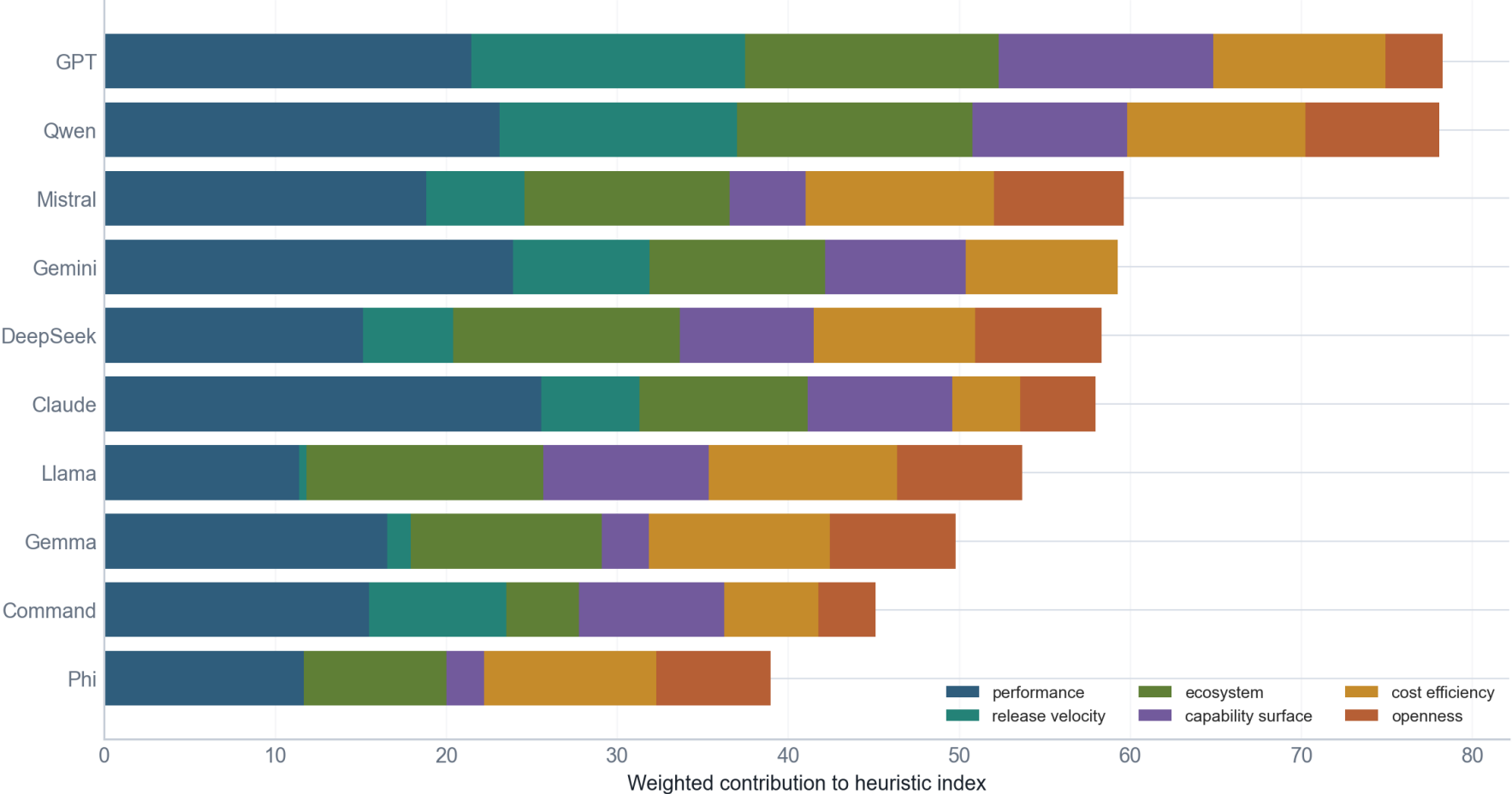


Stable top-tier families remain near the top across sensitivity variants; weight-sensitive families move materially when price or ecosystem weights change.

Company frontier scores

The headline rank is only the entry point. The stacked component chart below shows why a family ranks where it ranks. That matters because two families can have similar headline indexes for very different reasons: one may be performance-heavy, another may be ecosystem-heavy or cost-efficient.

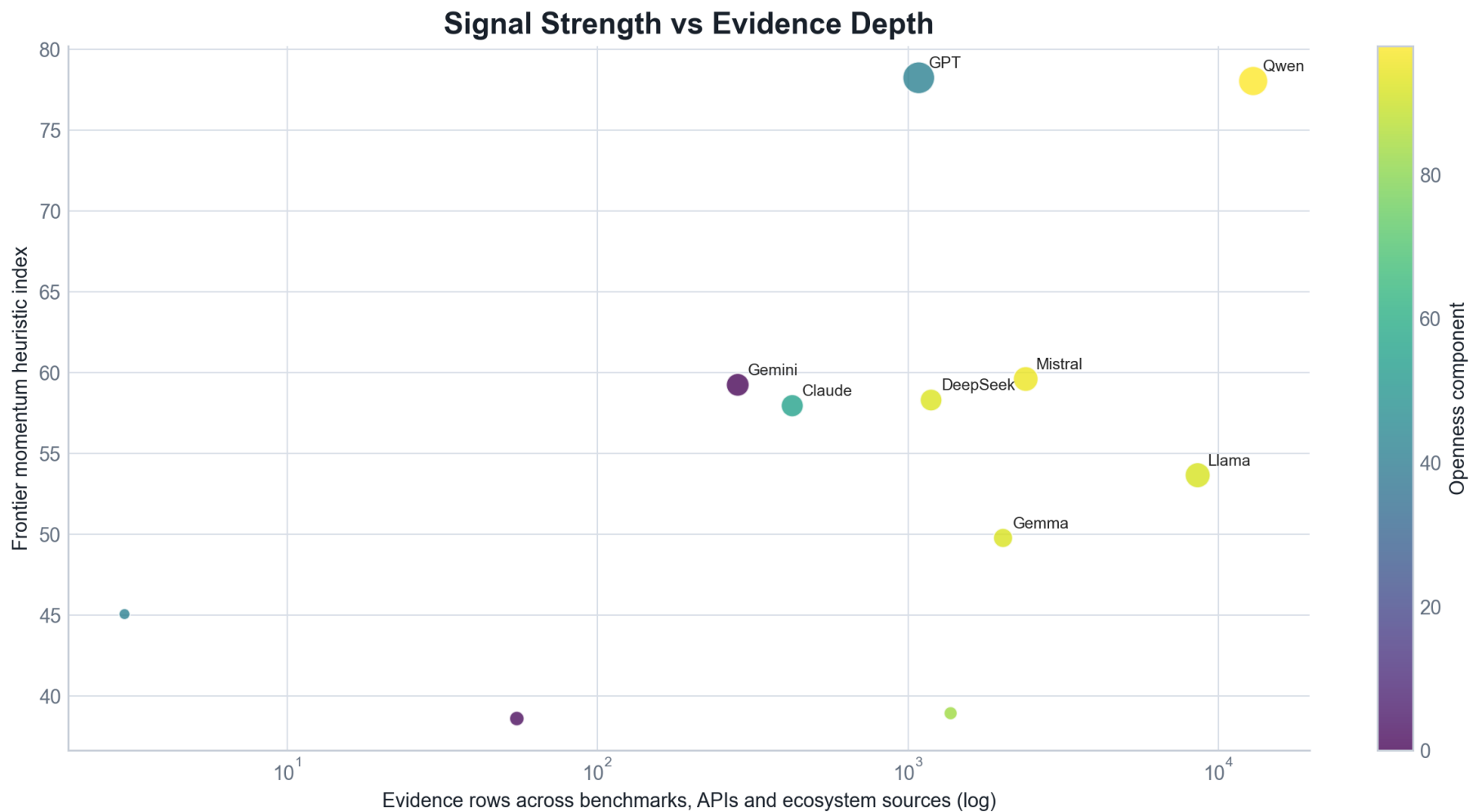
What Drives Each Frontier-Family Score



The stacked bars show contribution after baseline weights. They make the ranking auditable: a high score can come from benchmark performance, ecosystem pull, cost efficiency, or openness.

Score component stack

The evidence-depth scatter is the reviewer sanity check. A family with high score and high evidence count is more defensible than a family with a high score from sparse rows. Bubble size is tied to API catalog breadth, while color shows openness.



Bigger dots indicate more API catalog rows. This chart separates strong scores backed by broad evidence from sparse but visually impressive outliers.

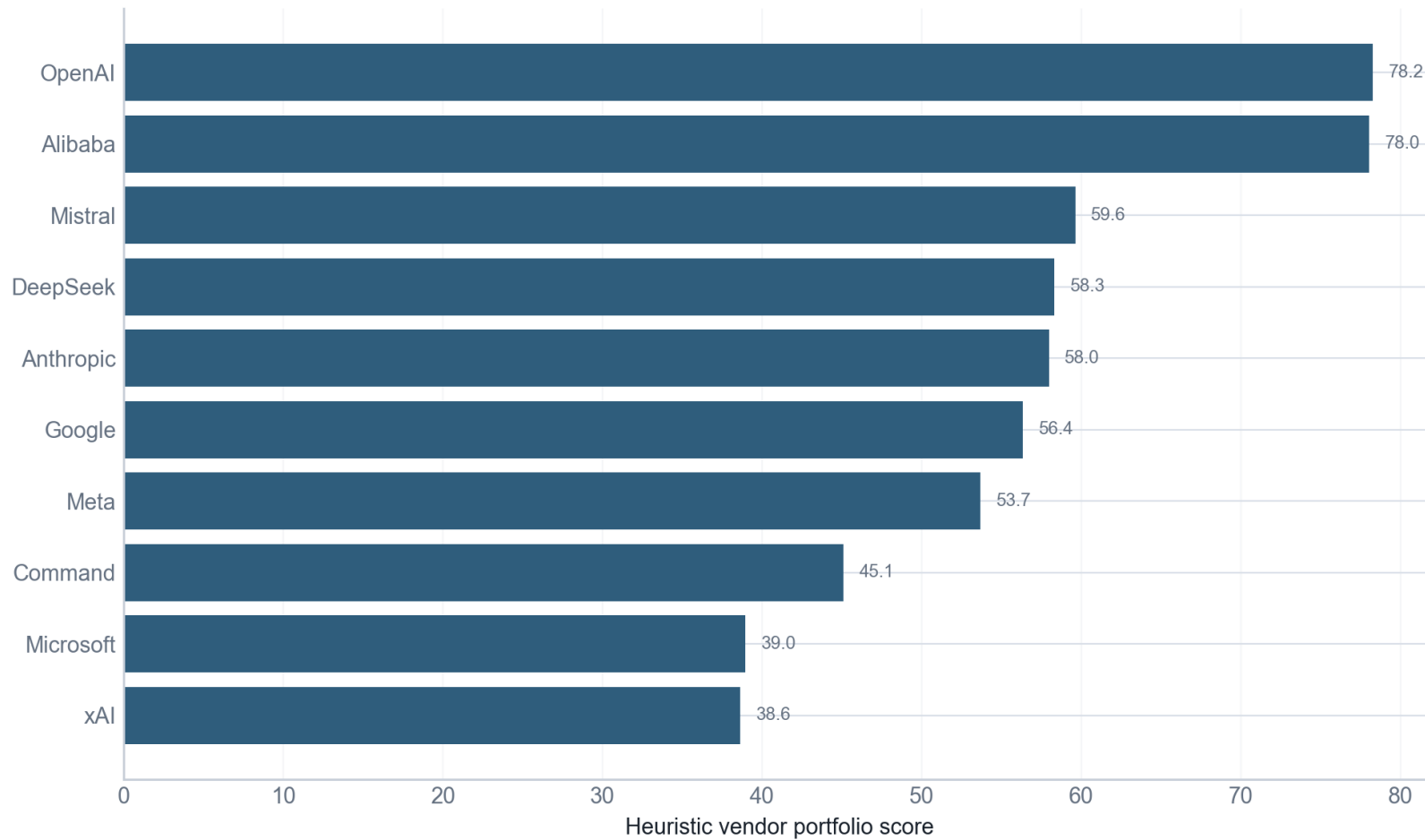
Score evidence scatter

Family Ranking vs Vendor Portfolio Ranking

Reviewers often reason in terms of companies, but model families remain the cleaner technical unit. The vendor view is therefore a companion view: it blends the flagship family with the evidence-weighted portfolio mean and keeps the flagship family visible.

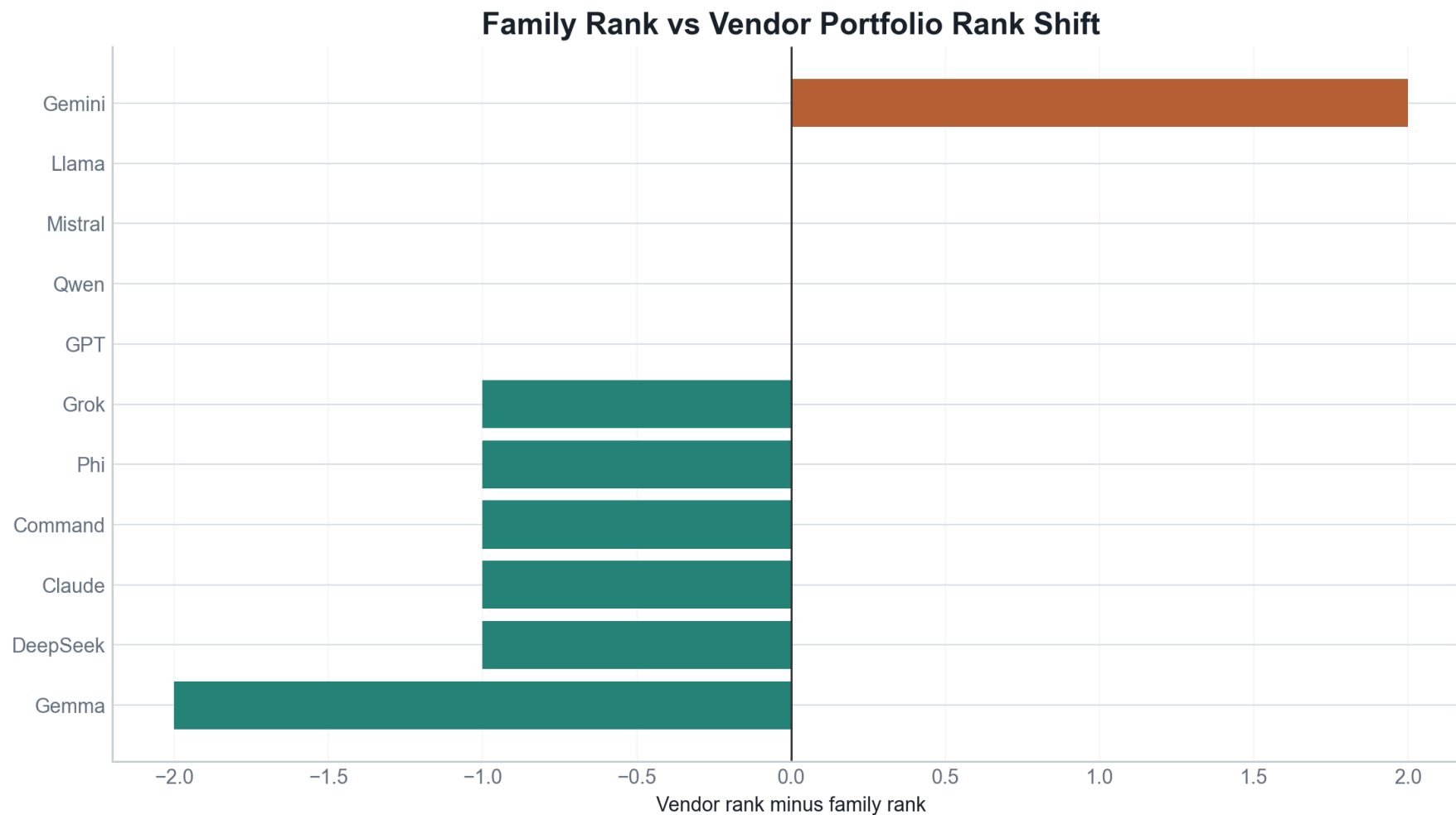
rank	vendor	vendor_frontier_portfolio_score	flagship_family	family_count	portfolio_families	evidence_count
1	OpenAI	78.25	GPT	1	GPT	1083
2	Alibaba	78.05	Qwen	1	Qwen	12932
3	Mistral	59.62	Mistral	1	Mistral	2394
4	DeepSeek	58.32	DeepSeek	1	DeepSeek	1187
5	Anthropic	57.97	Claude	1	Claude	424
6	Google	56.35	Gemini	2	Gemini,Gemma	2307
7	Meta	53.68	Llama	1	Llama	8569
8	Command	45.08	Command	1	Command	3
9	Microsoft	38.95	Phi	1	Phi	1372
10	xAI	38.63	Grok	1	Grok	55

Vendor Frontier Portfolio Score



Vendor score combines flagship family signal with evidence-weighted portfolio breadth. It is a business-facing companion to family rank, not a replacement.

Vendor frontier scores



Negative values mean the vendor portfolio rank is stronger than the individual family rank; positive values mean the opposite.

Family vs vendor rank shift

Who Builds The Next Best Model?

This table is not a prediction market. It is a Monte Carlo stress test over the scoring components: benchmark performance, release velocity, ecosystem pull, capability surface, cost and openness. `simulation_win_share` is the share of simulation draws won by each family, not a calibrated real-world probability. The corrected version separates frontier-quality leadership from open-ecosystem upside. The former asks who is most likely to make the raw best model; the latter asks who benefits if distribution and low cost matter more.

2-year simulated leaders:

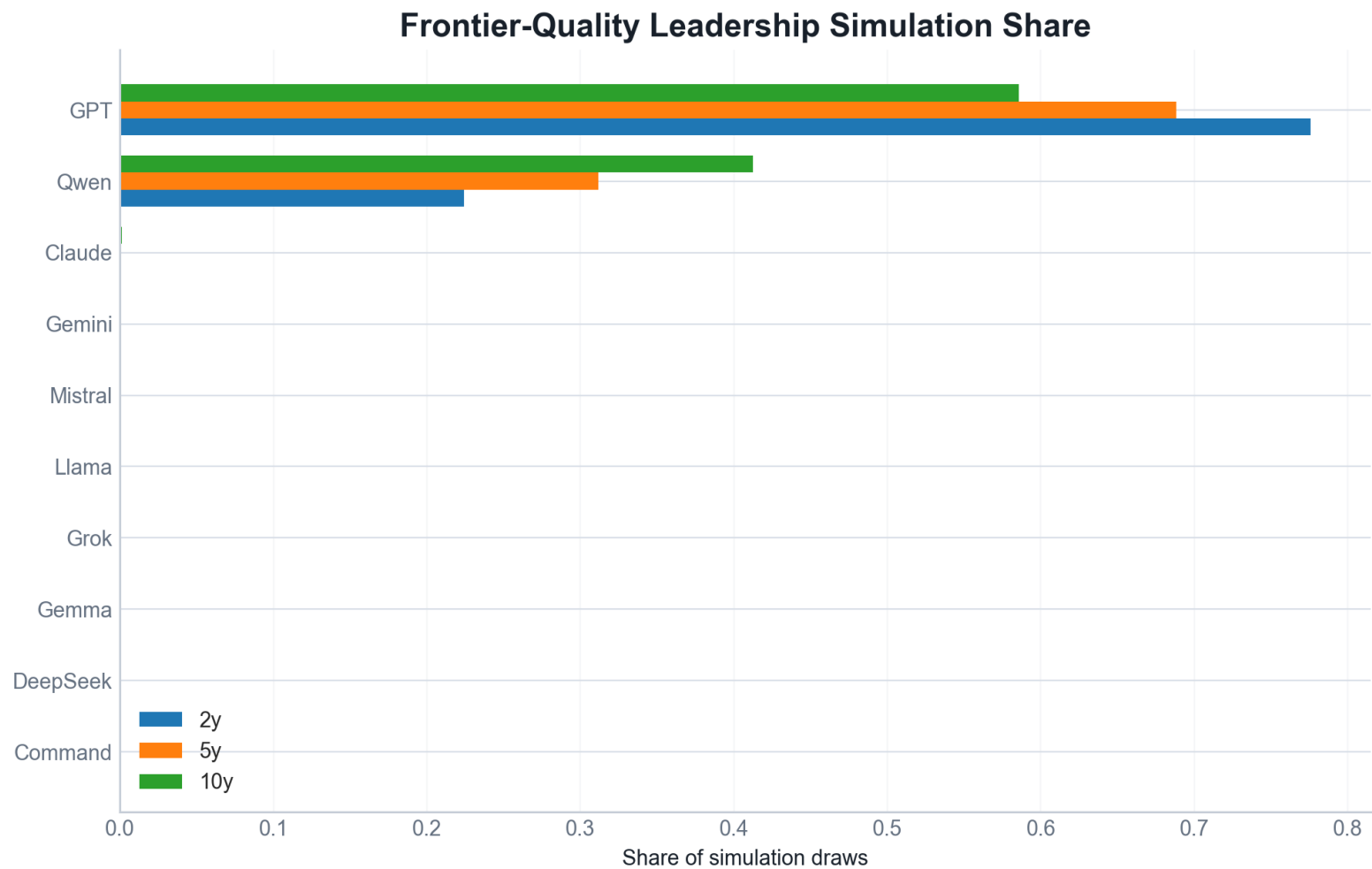
model_family	simulation_win_share	simulated_score_p10	simulated_score_p90
GPT	77.6%	73.85	81.7
Qwen	22.4%	70.51	78.46
Mistral	0.0%	46.59	55
Gemini	0.0%	57.91	66.11
DeepSeek	0.0%	44.88	52.81
Claude	0.0%	57.86	66.27
Llama	0.0%	34.67	43.91
Gemma	0.0%	34.28	43.5
Command	0.0%	44.26	51.65
Phi	0.0%	23.28	31.95

10-year simulated leaders, frontier-quality scenario:

model_family	simulation_win_share	simulated_score_p10	simulated_score_p90
GPT	58.6%	70.99	82.47
Qwen	41.2%	69.61	80.94
Claude	0.1%	54.44	66.16
Gemini	0.1%	52.83	64.78
Mistral	0.0%	47.67	59.78
DeepSeek	0.0%	47.2	58.95
Llama	0.0%	40.13	52.59
Gemma	0.0%	36.67	49.24
Command	0.0%	41.1	52.46
Phi	0.0%	25.99	38.3

10-year simulated leaders, open-ecosystem-upside scenario:

model_family	simulation_win_share	simulated_score_p10	simulated_score_p90
Qwen	70.2%	74.56	85.67
GPT	29.7%	70.94	82.55
DeepSeek	0.1%	56.83	68.62
Mistral	0.1%	57.29	69.44
Llama	0.0%	52.72	65.69
Gemini	0.0%	48.71	61.04
Claude	0.0%	51.53	62.88
Gemma	0.0%	48.19	60.54
Command	0.0%	37.52	48.51
Phi	0.0%	36.84	49.38

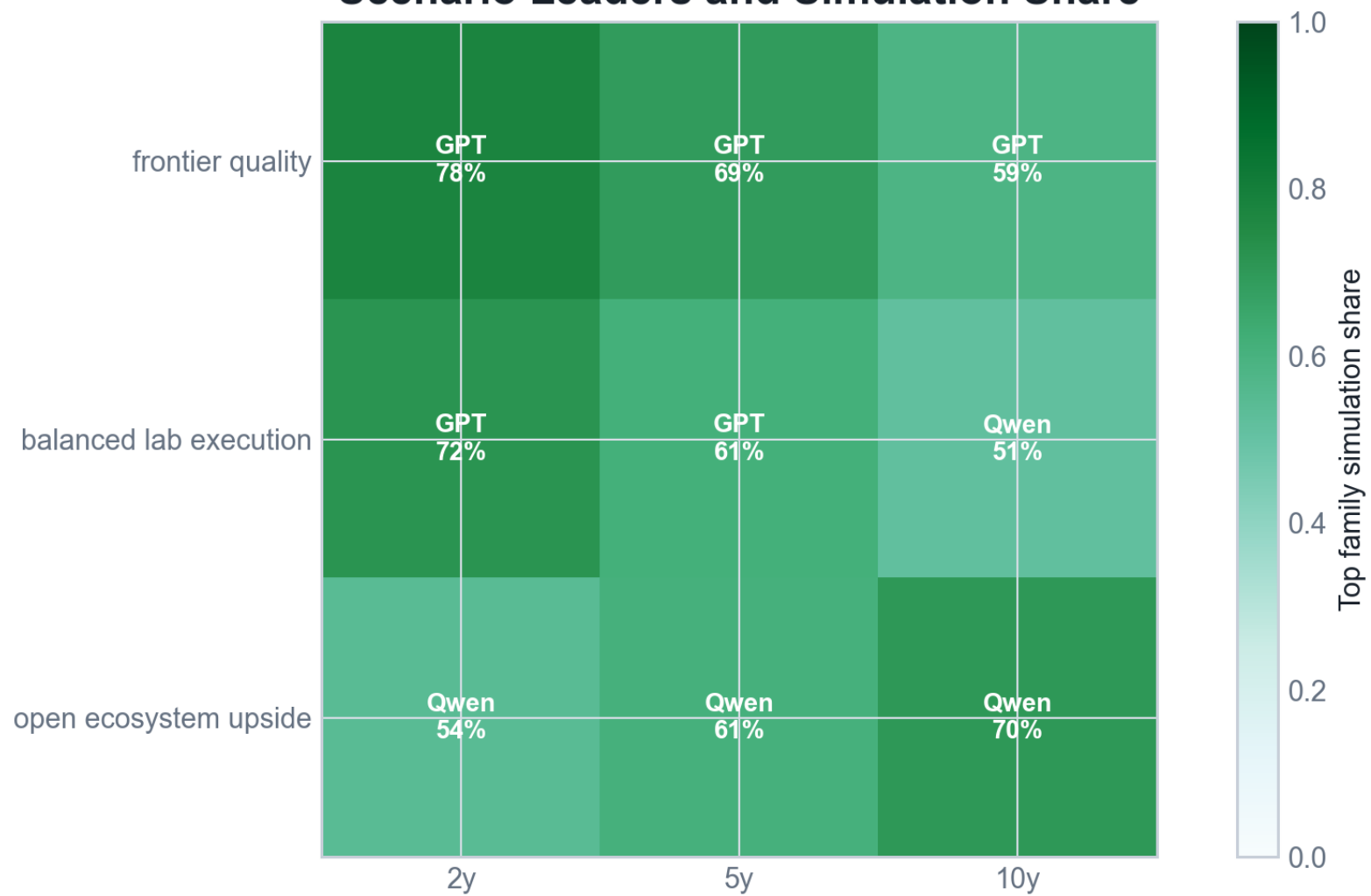


These are simulation shares under explicit component-weight scenarios. They are useful for stress-testing assumptions, not forecasting market odds.

Next frontier probabilities

The scenario matrix compresses the same simulation into a reviewer-friendly view: each cell names the leading family under a scenario/horizon pair and reports its share of simulation draws. This makes it obvious when the answer changes because the question changed.

Scenario Leaders and Simulation Share

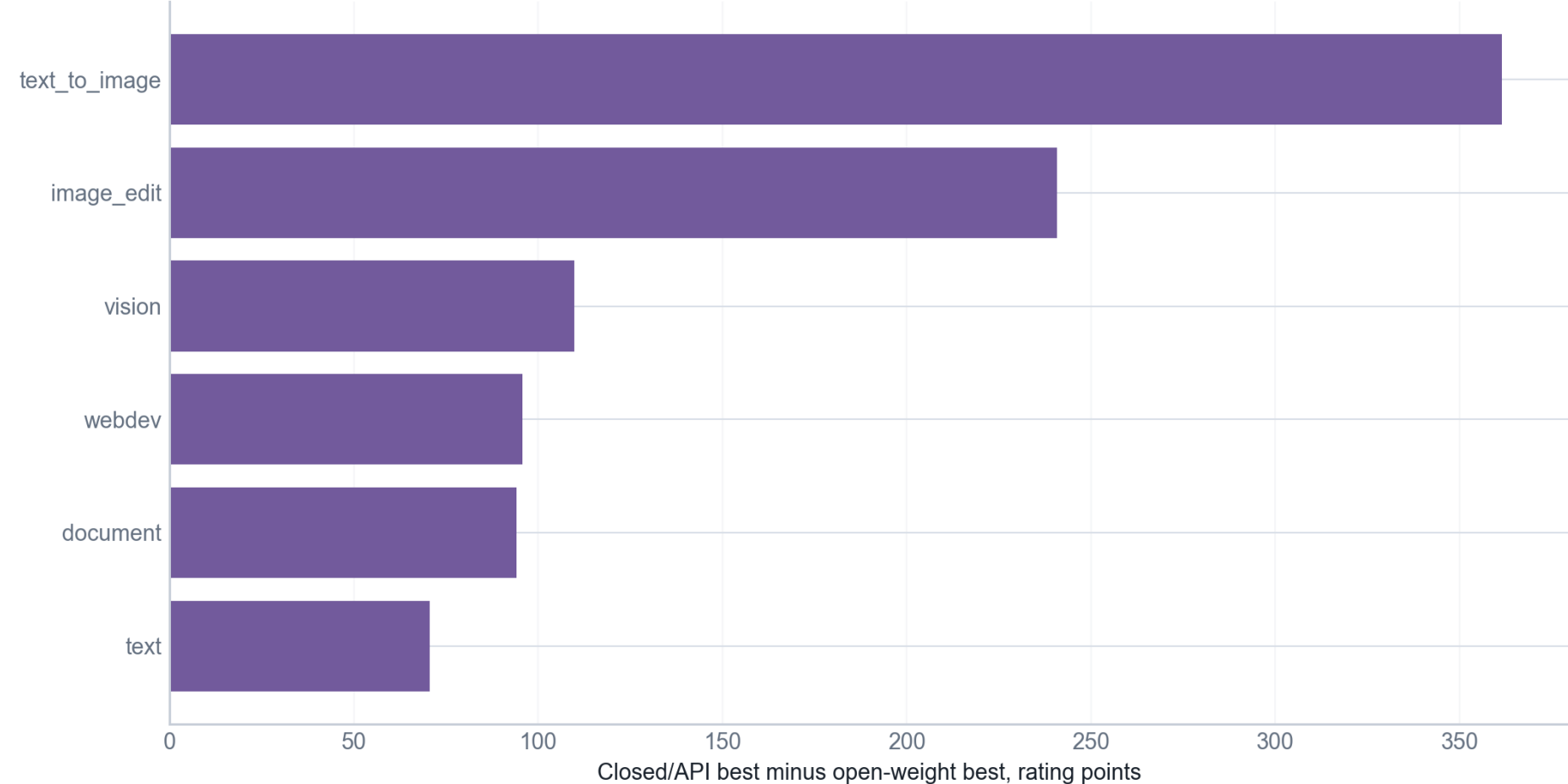


Leadership scenario matrix

Open vs Closed: Where Is The Gap?

category	closed_or_api	open_weight	open_closed_best_gap	open_closed_gap_pct_of_closed	comparison_note
text_to_image	1574.24	1212.6857458083412	361.56	0.2297	Comparable open-weight and closed/API rows observed in selected snapshot.
image_edit	1513.01	1272.2820948867309	240.73	0.1591	Comparable open-weight and closed/API rows observed in selected snapshot.
vision	1451.69	1341.839120952578	109.85	0.0757	Comparable open-weight and closed/API rows observed in selected snapshot.
webdev	1586.93	1491.3053126020395	95.62	0.0603	Comparable open-weight and closed/API rows observed in selected snapshot.
document	1527.83	1433.687985154499	94.14	0.0616	Comparable open-weight and closed/API rows observed in selected snapshot.
text	1619.78	1549.227871094854	70.55	0.0436	Comparable open-weight and closed/API rows observed in selected snapshot.
search	1255.86	n/a	n/a	0	No comparable open-weight or closed/API row in selected snapshot.

Open vs Closed LMArena Best-Model Gap by Category

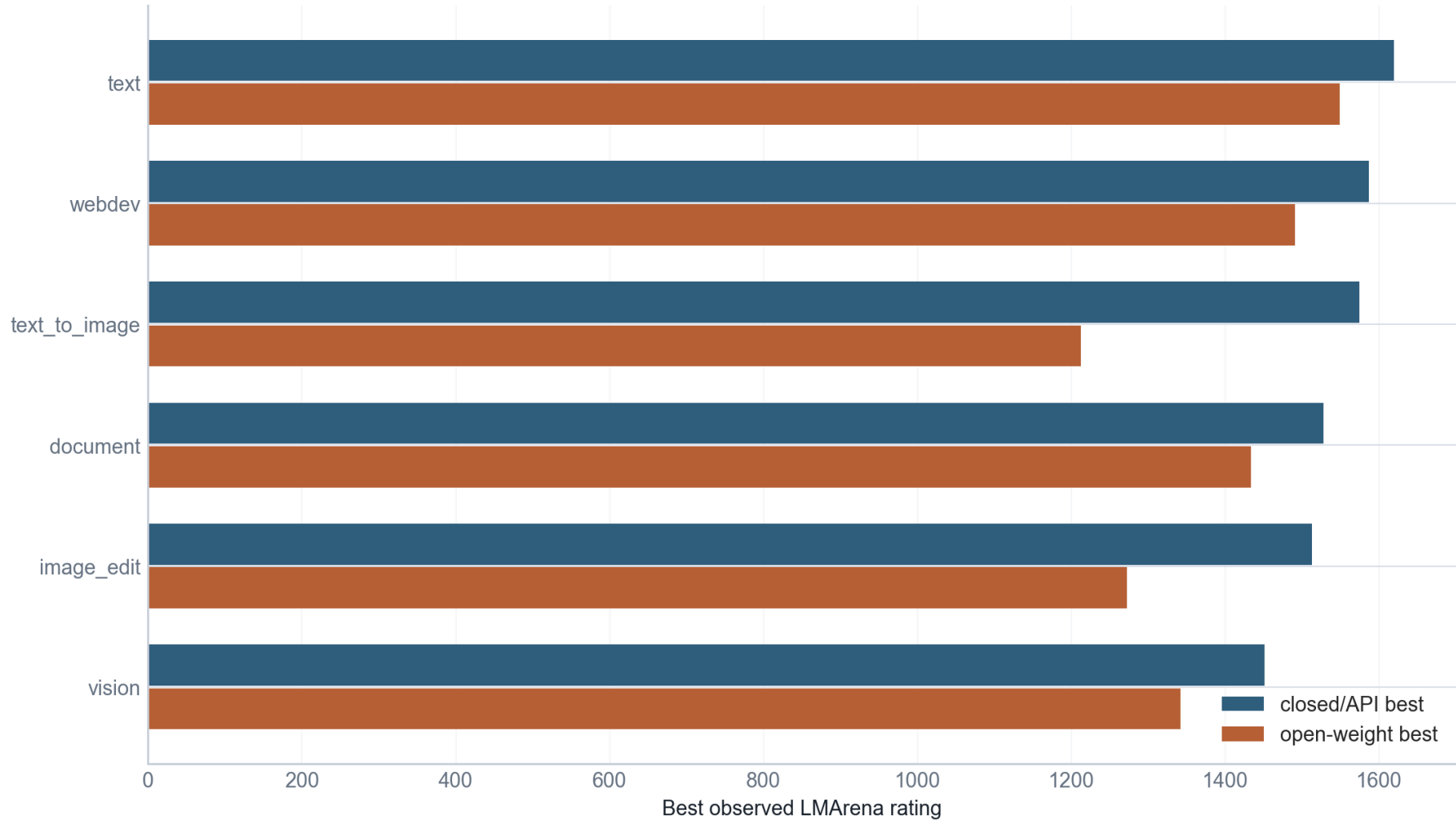


Categories without a comparable open-weight row are excluded here and called out in the table.

Open closed gap by category

The gap chart shows differences, but differences alone can hide whether both sides are high-quality. The paired rating chart below shows the actual open and closed best observed ratings by category where both sides exist.

Open and Closed Best Ratings by Category

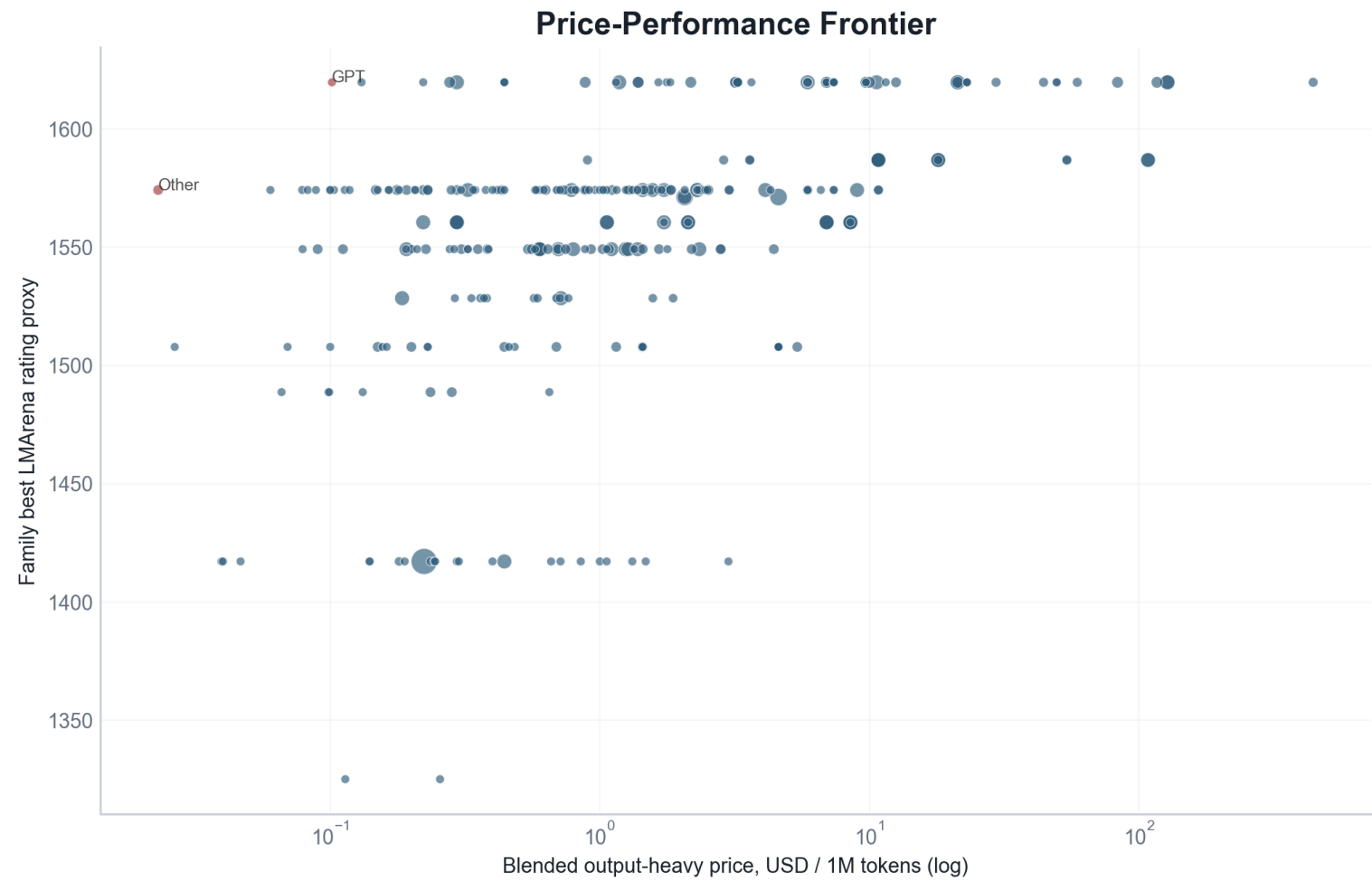


Open closed category levels

Price-Performance Frontier

Raw best model and economically deployable model are not the same decision. This frontier is explicitly a family-level proxy: OpenRouter model prices are joined to the best observed LMArena rating for the model family, not to a direct benchmark for every listed model. It should be read as a deployability screen, not model-level proof.

canonical_model	model_family	access_class	blended_price_usd_per_1m	family_best_lmarena	quality_proxy_level	price_performance_index
OpenAI: gpt-oss-20b	GPT	open_weight	0.1015	1619.78	family_level_proxy	91.48
inclusionAI: Ling-2.6-flash	Other	unknown	0.023	1574.24	family_level_proxy	82.18



Red points are non-dominated by this family-level proxy. Bubble size tracks context window, so cheap long-context models stand out without pretending they have direct model-level benchmark ratings.

Price performance frontier

The context-price map keeps three product dimensions visible at once: context window, blended token price and family-level rating proxy. It prevents a common mistake in AI market analysis: treating cheap, long-context and high-quality as one metric.



This view shows why context length, price and rating proxy must be kept separate: large context is a product surface, not a benchmark score.

Context price rating map

Direct Model Evidence vs Family Proxy

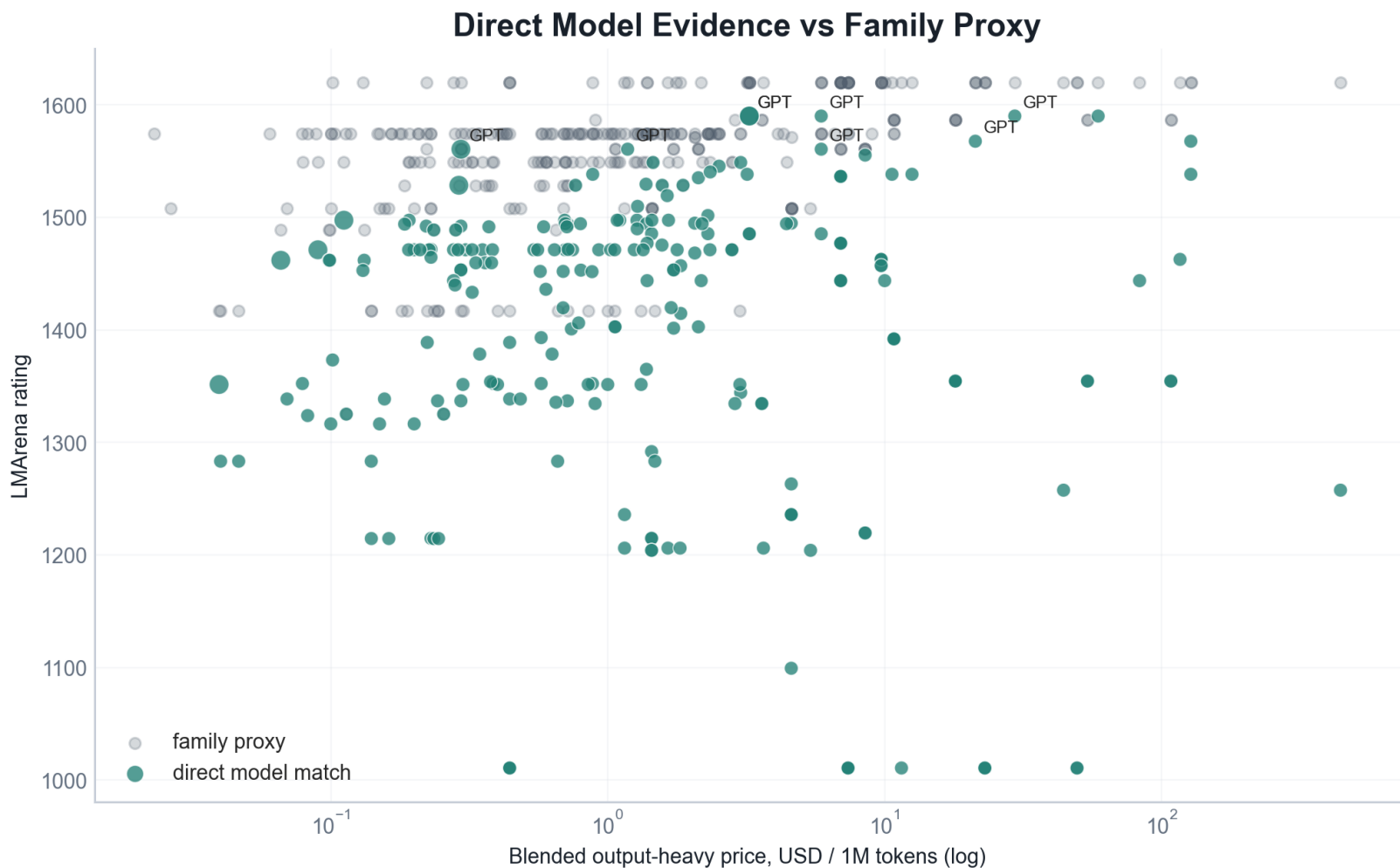
The audit table matches OpenRouter model IDs/names to LMArena, SWE-bench, LiveBench and Open LLM Leaderboard rows. Direct model matches are separated from family_only evidence so the deployability screen does not quietly inherit model-level certainty it does not have.

Match confidence audit:

match_confidence	rows
family_only	813
normalized_exact	504
unmatched	91

Direct evidence price-performance rows:

canonical_model	model_family	blended_price_usd_per_1m	direct_evidence_sources	direct_lmarena_rating	direct_lmarena_match_confidence	direct_price_performance_index	quality_proxy_level
OpenAI: GPT-4.1 Nano	GPT	0.295	lmarena	1560.56	normalized_exact	92.21	direct_model_lmarena
OpenAI: GPT-4.1 Mini	GPT	1.18	lmarena	1560.56	normalized_exact	91.18	direct_model_lmarena
OpenAI: o3 Mini High	GPT	3.245	livebench,lmarena,swebench	1590.01	normalized_exact	91.17	direct_model_lmarena
OpenAI: o3 Mini	GPT	3.245	livebench,lmarena,swebench	1590.01	normalized_exact	91.17	direct_model_lmarena
OpenAI: o3	GPT	5.9	livebench,lmarena,swebench	1590.01	normalized_exact	90.21	direct_model_lmarena
OpenAI: GPT-4.1	GPT	5.9	lmarena	1560.56	normalized_exact	88.9	direct_model_lmarena
OpenAI: GPT-5.5	GPT	21.25	lmarena	1567.64	normalized_exact	87.52	direct_model_lmarena
OpenAI: o3 Deep Research	GPT	29.5	livebench,lmarena,swebench	1590.01	normalized_exact	87.28	direct_model_lmarena
Google: Gemini 3 Flash Preview	Gemini	2.125	lmarena	1535.16	normalized_exact	87.13	direct_model_lmarena
OpenAI: GPT-5.4 Nano	GPT	0.8825	lmarena	1538.27	normalized_exact	87.06	direct_model_lmarena
Z.ai: GLM 5	Other	1.458	lmarena	1548.75	normalized_exact	86.86	direct_model_lmarena
MoonshotAI: Kimi K2.6	Other	2.524	lmarena	1545.48	normalized_exact	86.12	direct_model_lmarena



Direct points require model-level name matches. Proxy points keep the broader deployability screen visible without upgrading family evidence to model-level proof.

Direct vs proxy price performance

LLM Cost Per Message vs Fixed Task Cost

This section separates two claims that are often blended together. A modeled average message/task can become more expensive when users route more work to long-context, tool-heavy or agentic frontier runs. A fixed task, such as thesis-quality long-form writing under a stable token budget and quality threshold, can become cheaper when cheaper families become good enough. The tables below do not claim to observe private invoices or usage logs; they expose the assumptions behind the workload mix and the fixed-task thresholds.

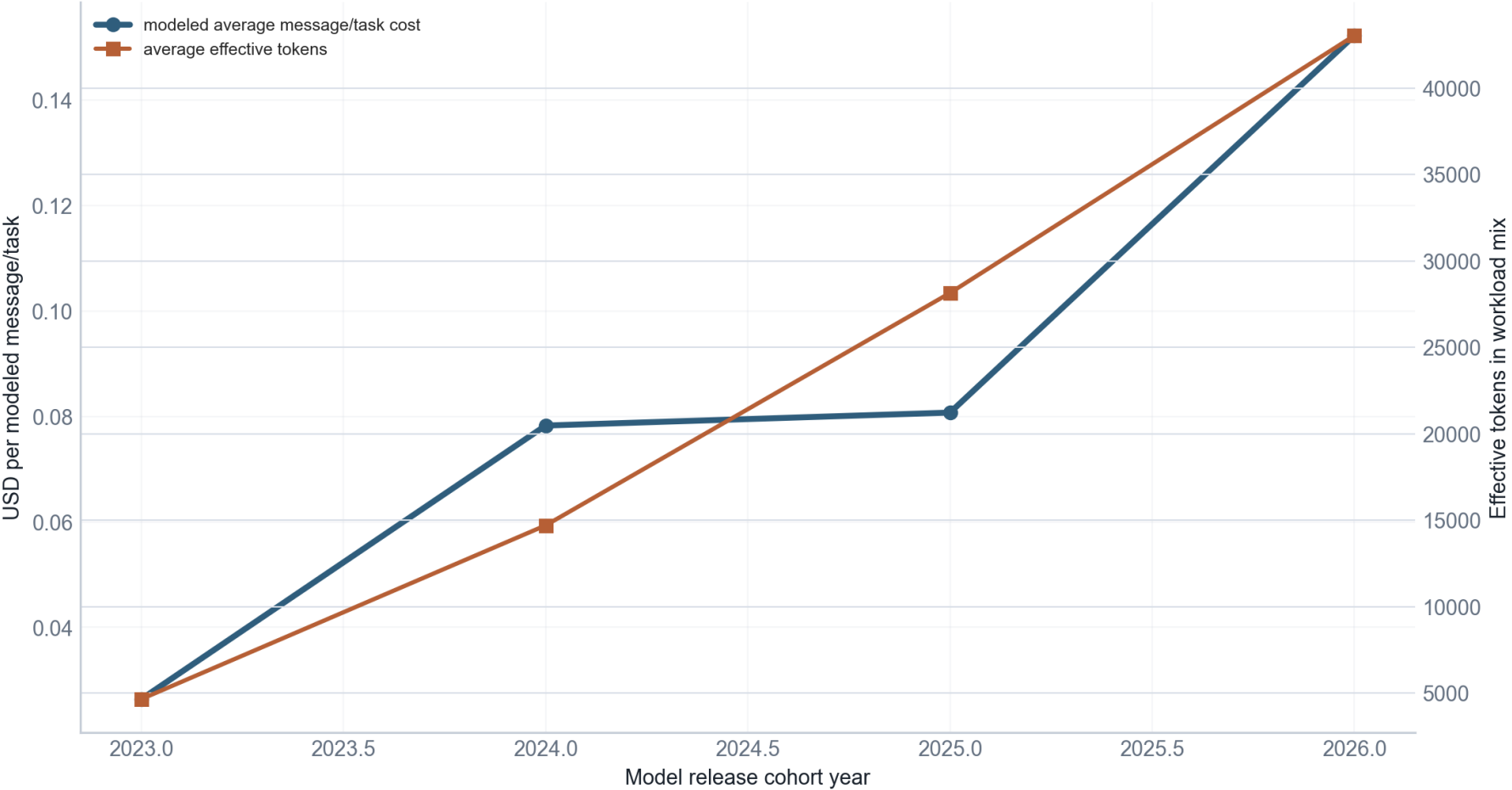
Modeled message/task cost by release cohort:

year	released_model_count	low_cost_blended_price_usd_per_1m	median_blended_price_usd_per_1m	frontier_blended_price_usd_per_1m	average_effective_tokens	modeled_average_message_cost_usd	message_cost_index_2023_100
2023	10	0.4788	1.4125	46.5	4655	0.026383	100
2024	57	0.209924	0.847	14.7	14708	0.078275	296.7
2025	232	0.21	0.8252	7.4125	28153	0.080733	306
2026	321	0.247	1.02	9.6	43046	0.152251	577.1

2026 workload profile components:

display_name	mix_share	input_tokens	output_tokens	price_quantile	profile_cost_usd	weighted_cost_contribution_usd
Simple chat or Q&A	0.32	900	500	0.2	0.000275	8.8e-05
Knowledge-work message	0.34	3500	1200	0.5	0.0032	0.001088
Long-context analysis	0.22	45000	5000	0.75	0.08125	0.017875
Agentic workflow run	0.12	220000	30000	0.9	1.11	0.1332

Modeled LLM Cost per Message/Task



This is a workload-mix model over listed-price cohorts, not observed billing data. The point is to separate per-token price from task complexity.

LLM message cost trends

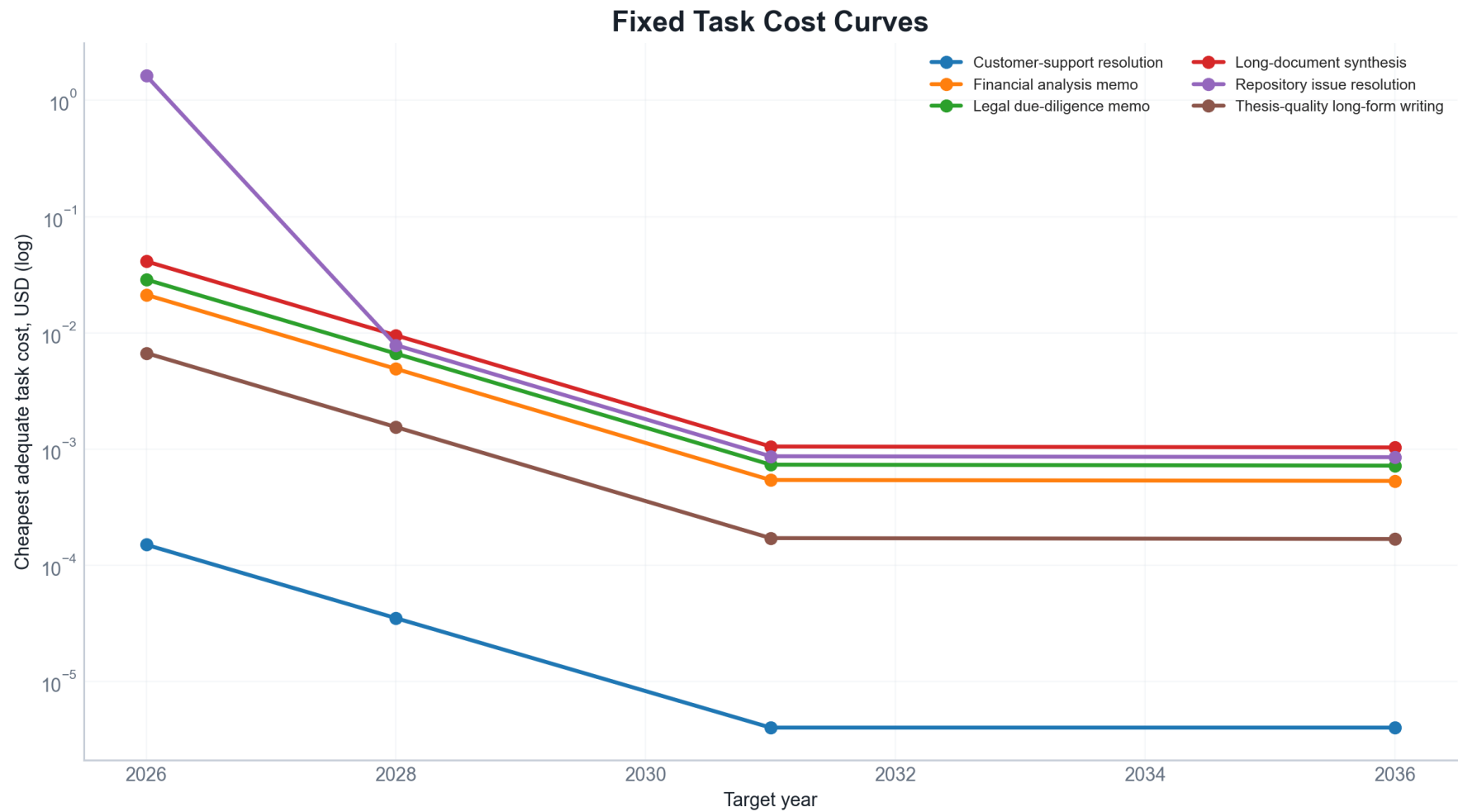
Current cheapest adequate fixed-task candidates:

display_name	domain_label	selected_model	selected_family	required_domain_score	selected_domain_score	forecast_task_cost_usd	adequacy_status	human_gate
Thesis-quality long-form writing	Language and writing	OpenAI: gpt-oss-20b	GPT	88	85.432	0.0067	best_available_below_threshold	advisor, fact and citation review
Repository issue resolution	Software engineering	Anthropic Claude Sonnet Latest	Claude	82	78.966	1.635	best_available_below_threshold	senior engineer review and tests
Financial analysis memo	Finance and quantitative analysis	Anthropic: Claude 3 Haiku	Claude	65	62.962	0.02125	best_available_below_threshold	assumption and control owner signoff
Legal due-diligence memo	Legal reasoning	Anthropic: Claude 3 Haiku	Claude	90	90.5	0.02875	adequate	licensed legal review

display_name	domain_label	selected_model	selected_family	required_domain_score	selected_domain_score	forecast_task_cost_usd	adequacy_status	human_gate
Long-document synthesis	Search and document work	Anthropic: Claude 3 Haiku	Claude	84	100	0.04125	adequate	source-grounding and factual review
Customer-support resolution	Instruction following	Meta: Llama 3.1 8B Instruct	Llama	78	81.208	0.00015	adequate	policy, refund and safety review

Base scenario fixed-task curves:

display_name	target_year	selected_family	selected_domain_score	forecast_task_cost_usd	cost_factor_vs_current	adequacy_status
Thesis-quality long-form writing	2028	GPT	96.11	0.001544	0.2304	adequate
Thesis-quality long-form writing	2031	GPT	98.854	0.000171	0.02548	adequate
Thesis-quality long-form writing	2036	GPT	98.854	0.000168	0.025	adequate
Repository issue resolution	2028	GPT	82.627	0.007834	0.004791	adequate
Repository issue resolution	2031	GPT	89.585	0.000866	0.00053	adequate
Repository issue resolution	2036	GPT	94.872	0.00085	0.00052	adequate
Financial analysis memo	2028	Claude	62.962	0.004896	0.2304	best_available_below_threshold
Financial analysis memo	2031	Claude	62.962	0.000541	0.02548	best_available_below_threshold
Financial analysis memo	2036	Claude	62.962	0.000531	0.025	best_available_below_threshold
Legal due-diligence memo	2028	Claude	90.5	0.006624	0.2304	adequate
Legal due-diligence memo	2031	Claude	90.5	0.000733	0.02548	adequate
Legal due-diligence memo	2036	Claude	90.5	0.000719	0.025	adequate
Long-document synthesis	2028	Claude	99.5	0.009504	0.2304	adequate
Long-document synthesis	2031	Claude	99.5	0.001051	0.02548	adequate
Long-document synthesis	2036	Claude	99.5	0.001031	0.025	adequate
Customer-support resolution	2028	Llama	81.208	3.5e-05	0.2304	adequate
Customer-support resolution	2031	Llama	81.208	4e-06	0.02548	adequate
Customer-support resolution	2036	Llama	81.208	4e-06	0.025	adequate



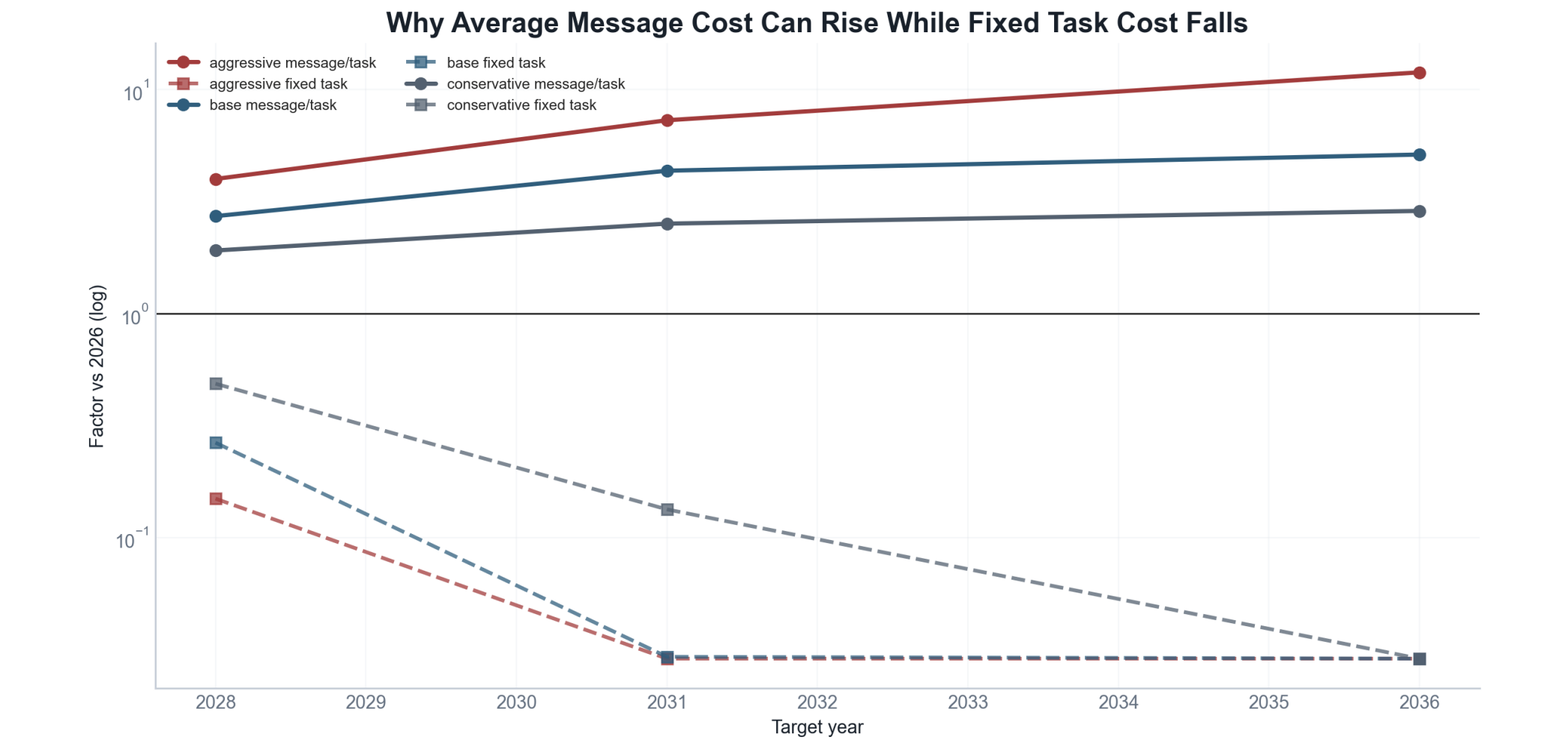
A fixed task can get cheaper when more families clear the required capability threshold and quality-adjusted unit cost falls.

Fixed task cost curves

The divergence table is the explicit version of the user's hypothesis: frontier work-unit cost can rise because average tasks get harder, while fixed task cost can fall because capability diffuses into cheaper models.

scenario	target_year	modeled_average_message_cost_usd	message_cost_factor_vs_2026	median_fixed_task_cost_usd	fixed_task_cost_factor_vs_2026	frontier_workload_complexity_multiplier
conservative	2028	0.291059	1.9117	0.012147	0.48588	1.9905
conservative	2031	0.383646	2.5198	0.003336	0.13344	2.7877
conservative	2036	0.43726	2.872	0.000719	0.02876	3.5149

scenario	target_year	modeled_average_message_cost_usd	message_cost_factor_vs_2026	median_fixed_task_cost_usd	fixed_task_cost_factor_vs_2026	frontier_workload_complexity_multiplier
base	2028	0.414459	2.7222	0.006624	0.26496	2.7222
base	2031	0.661146	4.3425	0.000733	0.02932	4.3425
base	2036	0.780511	5.1265	0.000719	0.02876	5.1265
aggressive	2028	0.607352	3.9892	0.003726	0.14904	3.5503
aggressive	2031	1.1106	7.2946	0.000719	0.02876	5.4509
aggressive	2036	1.81779	11.9394	0.000719	0.02876	6.6669



Solid lines track the workload mix becoming harder; dashed lines track the cheapest adequate model for fixed task profiles.



Each point is a task-model candidate. Adequacy uses family-domain benchmark scores, so this remains a deployability screen rather than direct model proof.

Fixed task quality cost ladder

Cost evidence notes:

source_id	name	used_for	url
openrouter_models_api	OpenRouter Models API	Current model price, context-window and modality catalog fields.	https://openrouter.ai/docs/guides/overview/models
epoch_llm_inference_price_trends	Epoch AI LLM inference price trends	External prior that quality-adjusted inference prices can fall faster than raw frontier price lists.	https://epoch.ai/data-insights/llm-inference-price-trends/
price_of_progress_arxiv_2511_23455	The Price of Progress: Algorithmic Efficiency and the Falling Cost of AI Inference	Quality-adjusted fixed-task cost-decline prior.	https://arxiv.org/abs/2511.23455
agentic_token_consumption_arxiv_2604_22750	How Do AI Agents Spend Your Money?	Token-amplification caveat for agentic coding and multi-step workflows.	https://arxiv.org/abs/2604.22750

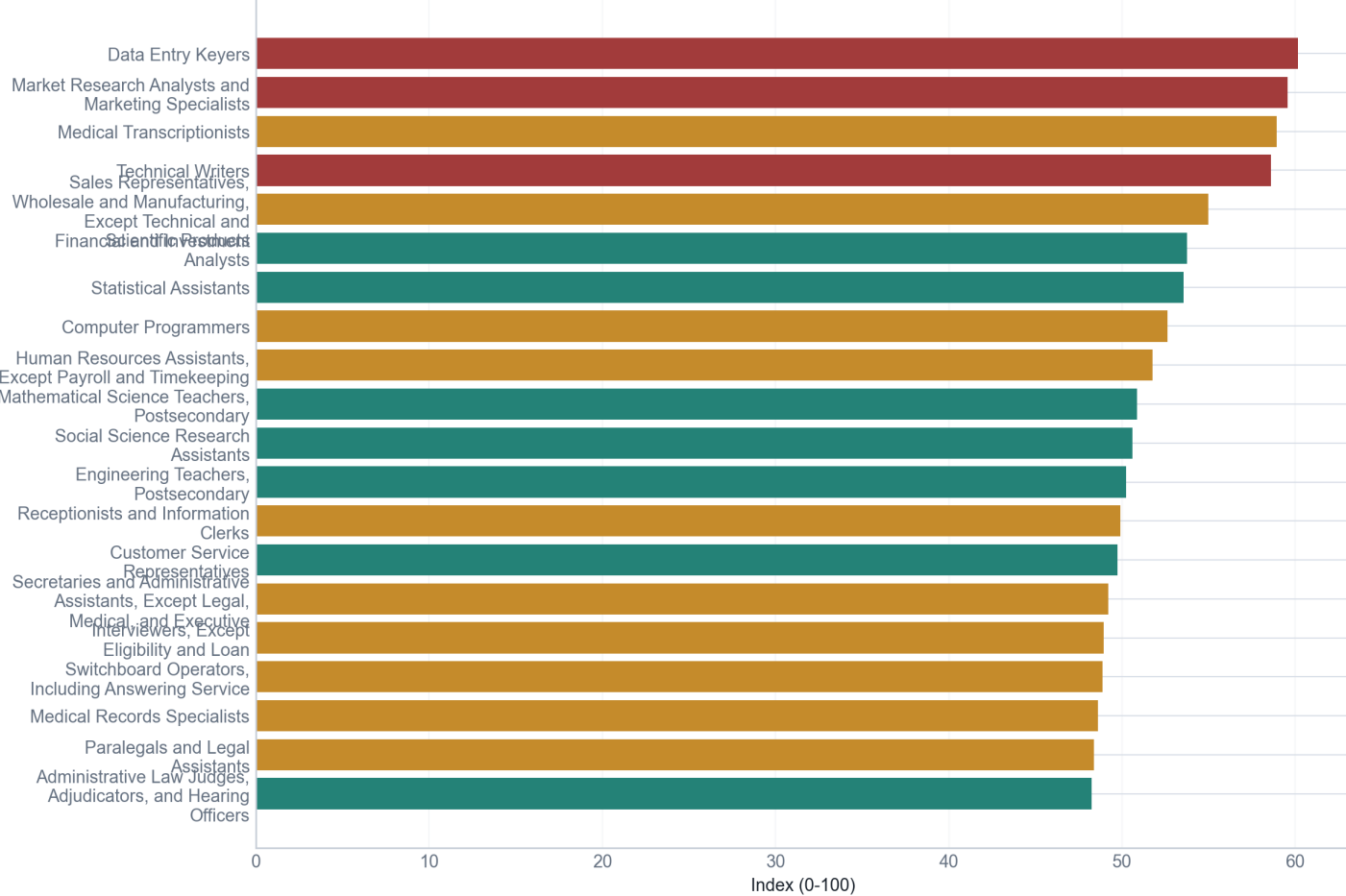
source_id	name	used_for	url
price_reversal_arxiv_2603_23971	The Price Reversal Phenomenon	Caveat that listed price can be a weak proxy for realized cost when thinking tokens and retry variance differ by model.	https://arxiv.org/abs/2603.23971
anthropic_economic_index_arxiv_2511_15080	Anthropic Economic Index report: Uneven geographic and enterprise AI adoption	External support for rising directive delegation and more autonomous AI task use.	https://arxiv.org/abs/2511.15080

Job Exposure And Labor Pressure

The labor table joins Anthropic observed occupation exposure to wage/job companion data, task-level penetration, automation/augmentation mode shares, and keyword-derived task bottlenecks from O*NET text. The output is an occupation-level pressure index, not a prediction that a whole occupation vanishes.

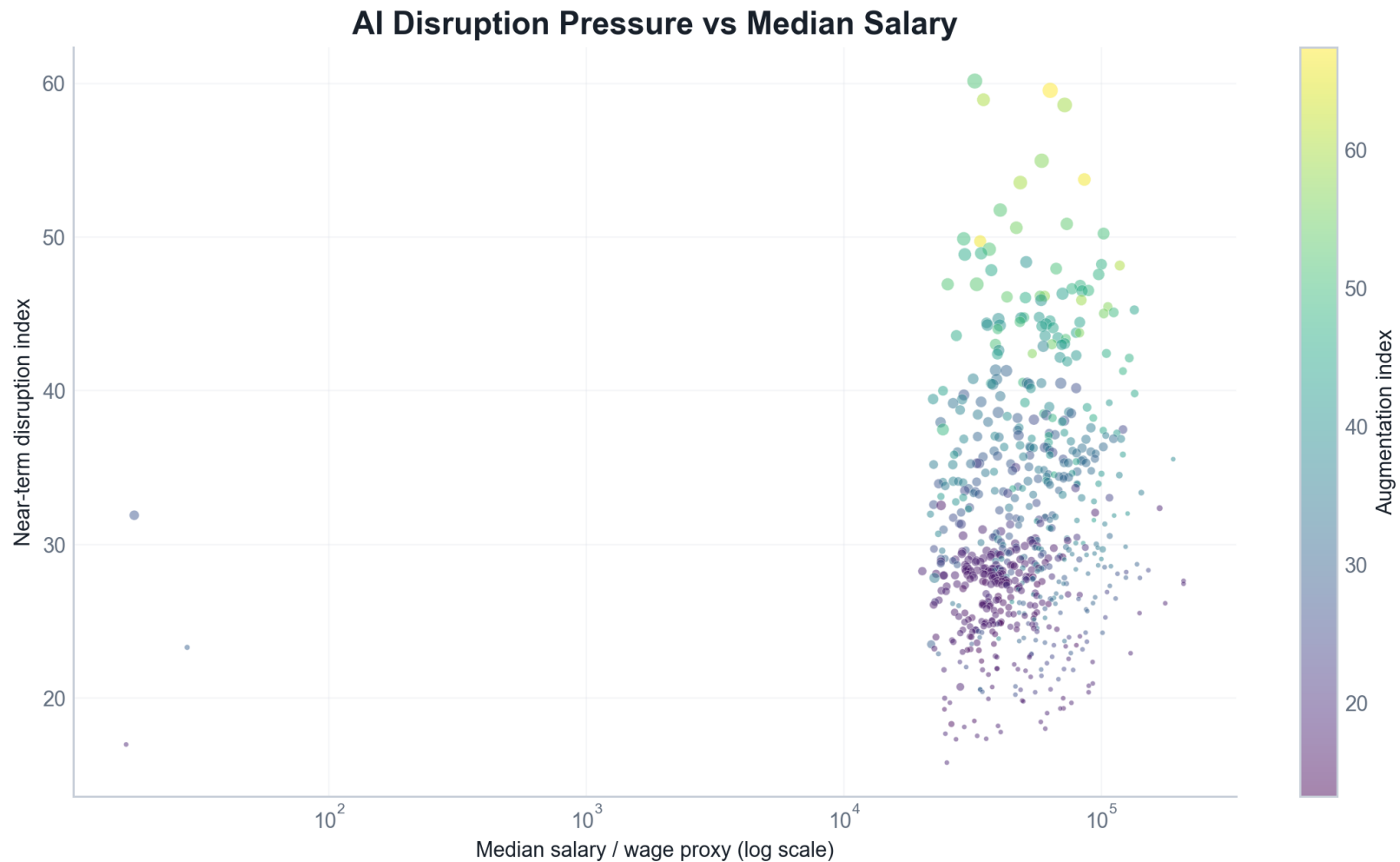
title	near_term_disruption_index	substitution_pressure_index	augmentation_index	full_job_automation_feasibility_index	dominant_outcome	risk_label
Data Entry Keyers	60.16	68.88	51.54	37.72	replacement_candidate	very_high
Market Research Analysts and Marketing Specialists	59.55	56.36	67.43	38.97	replacement_candidate	high
Medical Transcriptionists	58.94	60.44	60.15	28.27	mixed_redesign	high
Technical Writers	58.6	55.78	59.26	36.26	replacement_candidate	high
Sales Representatives, Wholesale and Manufacturing, Except Technical and Scientific Products	54.97	55.94	56.77	34.77	mixed_redesign	high
Financial and Investment Analysts	53.75	40.35	66.37	27.24	augmentation_first	high
Statistical Assistants	53.55	47.71	59.62	31.6	augmentation_first	high
Computer Programmers	52.61	52.14	55.26	26.07	mixed_redesign	high
Human Resources Assistants, Except Payroll and Timekeeping	51.76	50.06	52.03	30.44	mixed_redesign	high
Mathematical Science Teachers, Postsecondary	50.86	43.32	53.09	26.39	augmentation_first	high
Social Science Research Assistants	50.61	44.68	55.18	27.3	augmentation_first	high
Engineering Teachers, Postsecondary	50.23	40.29	50.32	24.37	augmentation_first	high

Highest Near-Term AI Disruption Pressure by Occupation



Color encodes the dominant modeled outcome, so the chart separates task disruption from whole-job replacement.

Job exposure top

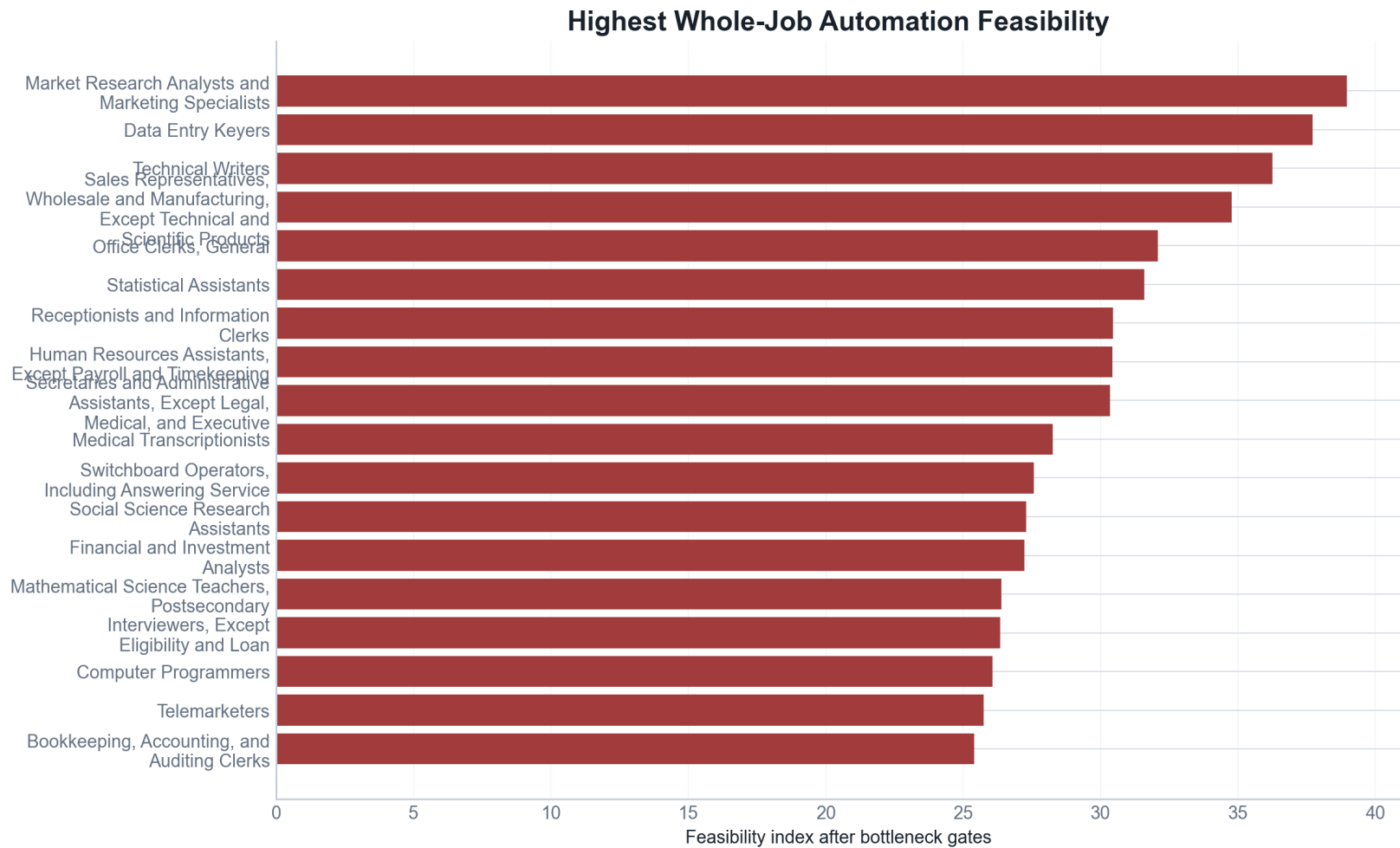


Wage scatter

Whole-Job Replacement Feasibility

The replacement feasibility index gates substitution pressure through physical, trust, regulatory and task-coverage bottlenecks. This is the section that answers the "will AI take all jobs" question more honestly: many occupations are touched; fewer are clean full-job replacement candidates.

title	job_family	full_job_automation_feasibility_index	substitution_pressure_index	human_bottleneck_index	dominant_outcome
Market Research Analysts and Marketing Specialists	Business and Financial Operations	38.97	56.36	0	replacement_candidate
Data Entry Keyers	Office and Administrative Support	37.72	68.88	0	replacement_candidate
Technical Writers	Arts, Design, Entertainment, Sports, and Media	36.26	55.78	0	replacement_candidate
Sales Representatives, Wholesale and Manufacturing, Except Technical and Scientific Products	Sales and Related	34.77	55.94	2.4	mixed_redesign
Office Clerks, General	Office and Administrative Support	32.08	51.56	1.22	mixed_redesign
Statistical Assistants	Office and Administrative Support	31.6	47.71	0.78	augmentation_first
Receptionists and Information Clerks	Office and Administrative Support	30.45	53.46	1.26	mixed_redesign
Human Resources Assistants, Except Payroll and Timekeeping	Office and Administrative Support	30.44	50.06	0	mixed_redesign
Secretaries and Administrative Assistants, Except Legal, Medical, and Executive	Office and Administrative Support	30.34	52.33	1.95	mixed_redesign
Medical Transcriptionists	Healthcare Support	28.27	60.44	9.95	mixed_redesign
Switchboard Operators, Including Answering Service	Office and Administrative Support	27.58	49.12	0.68	mixed_redesign
Social Science Research Assistants	Life, Physical, and Social Science	27.3	44.68	0.68	augmentation_first



This chart is intentionally stricter than task exposure: substitution pressure is gated by physical, trust, regulatory and task-coverage bottlenecks.

Replacement feasibility

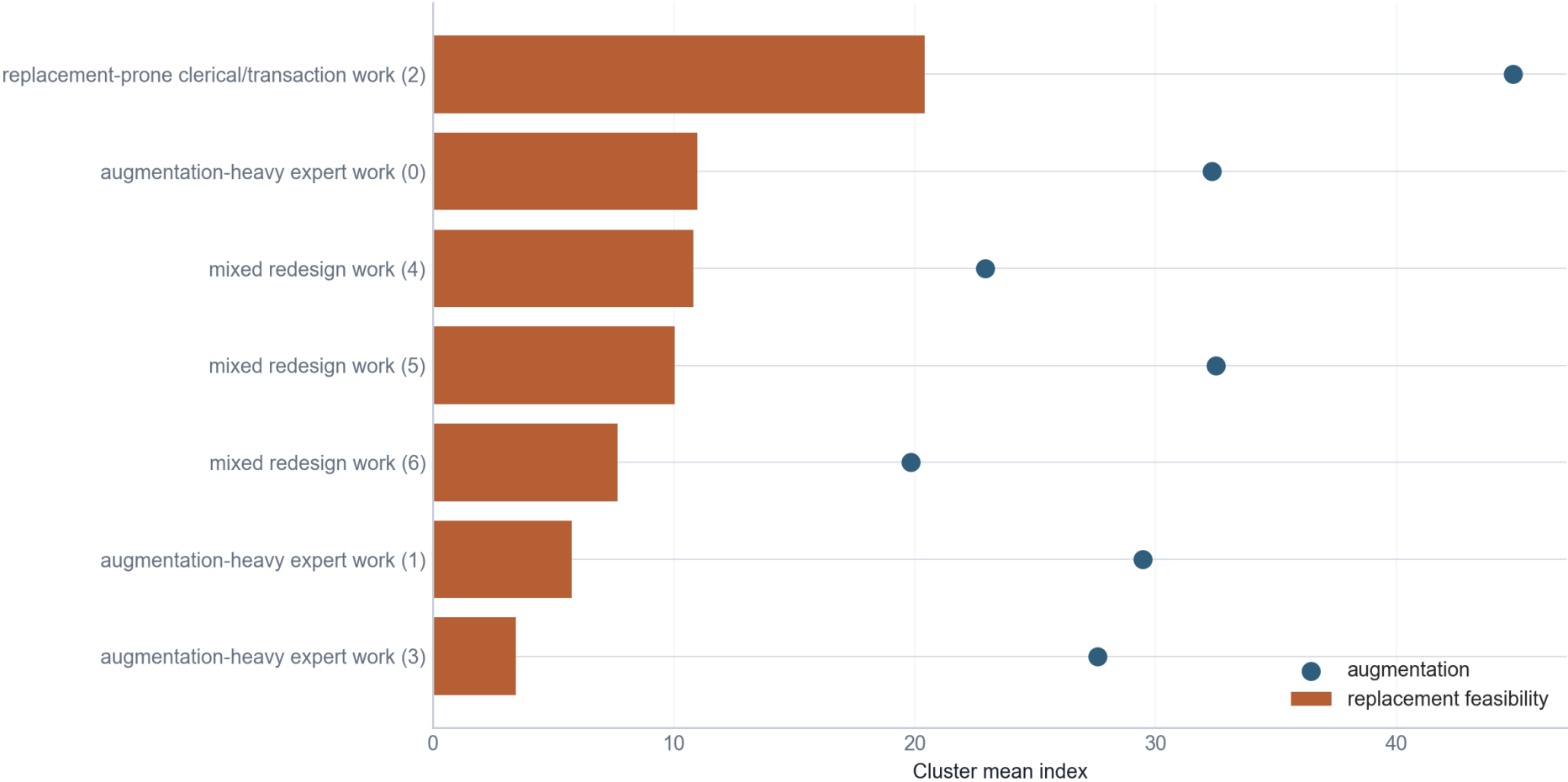
Labor Clusters

labor_cluster_id	cluster_label	occupation_count	full_job_automation_feasibility_index	augmentation_index	human_bottleneck_index	example_occupations
2	replacement-prone clerical/transaction work	113	20.4247	44.8353	1.31115	Data Entry Keyers; Market Research Analysts and Marketing Specialists; Technical Writers; Sales Representatives, Wholesale and Manufacturing, Except Technical and Scientific Products; Financial and Investment Analysts
0	augmentation-heavy expert work	28	10.9904	32.3325	1.56679	Desktop Publishers; Special Effects Artists and Animators; Photographic Process Workers and Processing Machine Operators; Prepress Technicians and Workers; Graphic Designers
4	mixed redesign work	298	10.799	22.9227	2.07715	Power Plant Operators; Baggage Porters and Bellhops; Hosts and Hostesses, Restaurant, Lounge, and Coffee Shop; Couriers and Messengers; Library Technicians
5	mixed redesign work	16	10.0306	32.4838	7.43938	Medical Transcriptionists; Medical Records Specialists; Medical Secretaries and Administrative Assistants; Judicial Law Clerks; Court Reporters and Simultaneous Captioners
6	mixed redesign work	104	7.67019	19.8389	7.65317	Insurance Appraisers, Auto Damage; Computer, Automated Teller, and Office Machine Repairers; Rail Yard Engineers, Dinkey Operators, and Hostlers; Amusement and Recreation Attendants; Bus and Truck Mechanics and Diesel Engine Specialists
1	augmentation-heavy expert work	152	5.75487	29.4653	2.2102	Title Examiners, Abstractors, and Searchers; Compensation and Benefits Managers; Magnetic Resonance Imaging Technologists; Production, Planning, and Expediting Clerks; Purchasing Managers
3	augmentation-heavy expert work	45	3.45956	27.58	9.87867	Nurse Practitioners; Nurse Midwives; Genetic Counselors; Special Education Teachers, Preschool; Nurse Anesthetists

Labor-weighted dominant outcome summary:

group	occupation_count	labor_weight_sum	weighted_disruption_index	weighted_replacement_feasibility	weighted_augmentation_index
replacement_candidate	3	22337.6	59.66	38.31	61.4
mixed_redesign	364	2.96894e+06	33.73	14.76	27.53
augmentation_first	389	4.21343e+06	32.69	9.76	36.22

Labor Clusters: Replacement Feasibility vs Augmentation

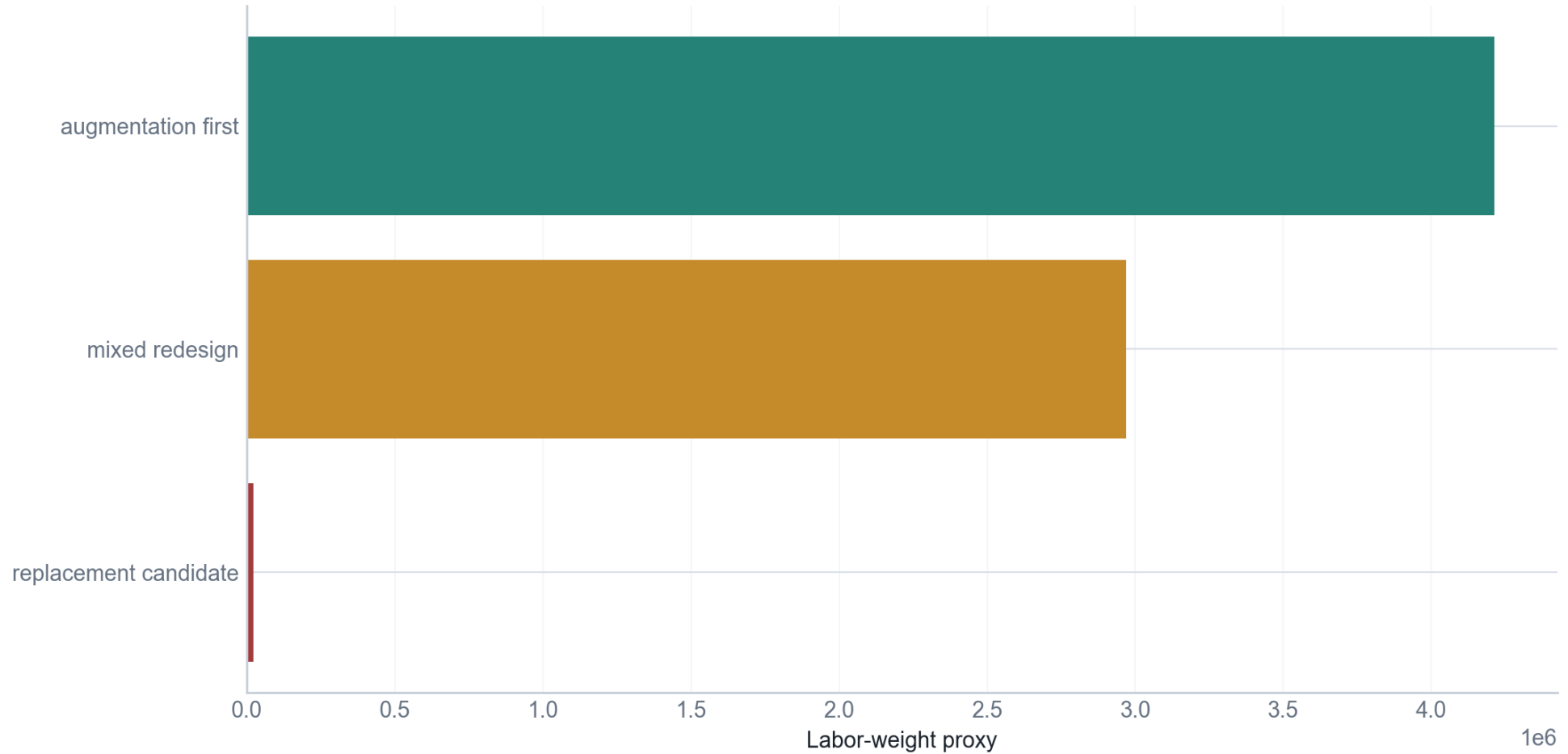


Bars and dots separate the two labor stories: replacement feasibility can stay lower than augmentation pressure even in exposed clusters.

Labor clusters

The labor-weighted outcome mix below is the report's guardrail against overclaiming. It weights modeled outcomes by the best available public labor proxy so a handful of highly automatable occupations do not dominate the narrative.

Labor-Weighted Dominant Outcome Mix



The weight is a public-data proxy, not an employment forecast. It helps keep the report from over-indexing on a few eye-catching occupations.

Labor outcome mix

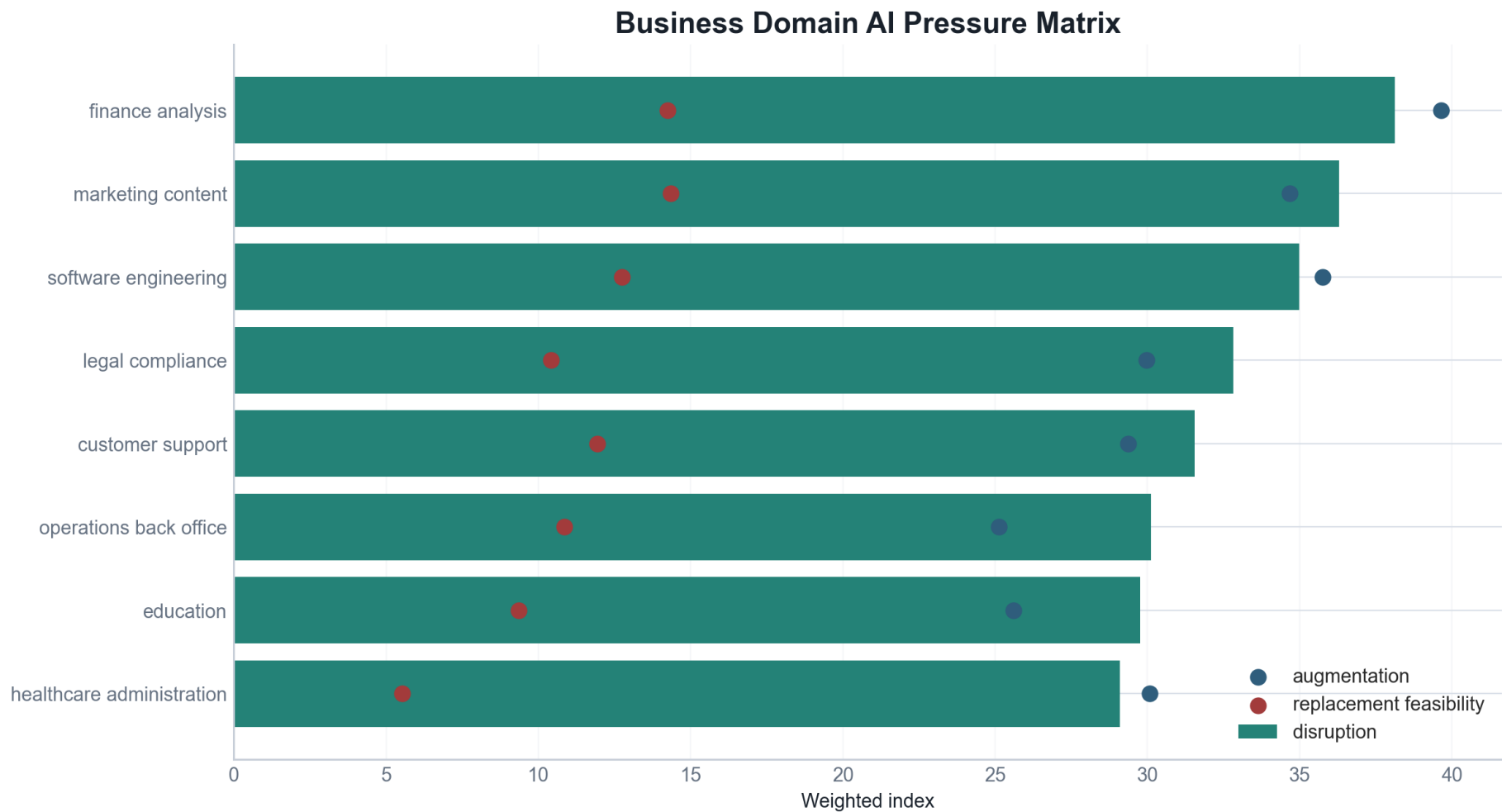
Business Domain Implications

The domain layer translates occupation-level pressure into business language. It does not replace occupation evidence; each domain keeps example occupations and explicit human gates so a portfolio reviewer can see where the abstraction could fail.

business_domain	pressure_label	disruption_index	augmentation_index	replacement_feasibility_index	human_bottleneck_index	example_occupations
finance_analysis	moderate	38.13	39.65	14.24	1.42	Market Research Analysts and Marketing Specialists; Financial and Investment Analysts; Accountants and Auditors; Loan Officers
marketing_content	moderate	36.29	34.67	14.34	1.21	Technical Writers; Public Relations Specialists; Real Estate Brokers; Real Estate Sales Agents
software_engineering	moderate	34.98	35.76	12.74	3.13	Computer Programmers; Social Science Research Assistants; Secretaries and Administrative Assistants, Except Legal, Medical, and Executive; Interviewers, Except Eligibility and Loan
legal_compliance	moderate	32.82	29.97	10.41	3.11	Paralegals and Legal Assistants; Administrative Law Judges, Adjudicators, and Hearing Officers; Judges, Magistrate Judges, and Magistrates; Property, Real Estate, and Community Association Managers
customer_support	moderate	31.55	29.36	11.93	3.11	Sales Representatives, Wholesale and Manufacturing, Except Technical and Scientific Products; Receptionists and Information Clerks; Customer Service Representatives; Switchboard Operators, Including Answering Service
operations_back_office	moderate	30.12	25.13	10.86	3.52	Data Entry Keyers; Statistical Assistants; Human Resources Assistants, Except Payroll and Timekeeping; Political Scientists
education	moderate	29.77	25.61	9.36	3.42	Mathematical Science Teachers, Postsecondary; Engineering Teachers, Postsecondary; English Language and Literature Teachers, Postsecondary; Health Specialties Teachers, Postsecondary
healthcare_administration	moderate	29.09	30.08	5.52	7.75	Medical Transcriptionists; Medical Records Specialists; Nurse Practitioners; Magnetic Resonance Imaging Technologists

Workflow examples:

business_domain	workflow_example	likely_ai_role	human_gate
software_engineering	Issue triage, code review, test generation and migration planning	draft patches and review checklists	senior engineer approval and production ownership
customer_support	Ticket summarization, answer drafting and escalation routing	suggest responses and detect repeated issues	human review for refunds, safety and account actions
legal_compliance	Contract review, policy comparison and evidence packet preparation	extract clauses and compare requirements	licensed professional signoff
marketing_content	Campaign briefs, SEO drafts, localization and asset variants	generate drafts and performance hypotheses	brand, legal and factual review
finance_analysis	Variance explanations, spreadsheet checks and memo drafting	surface anomalies and draft analysis	accountability for assumptions and controls
healthcare_administration	Claims coding, prior authorization packets and patient-message drafts	prepare structured documentation	clinical and compliance review
education	Lesson planning, tutoring support and feedback drafting	adapt materials and summarize progress	teacher judgment and student relationship
operations_back_office	Data entry cleanup, scheduling, reconciliation and process documentation	automate routine transformations	exception handling and vendor/customer accountability



The domain view translates occupation-level exposure into business language while keeping augmentation and replacement separate.

Business domain pressure matrix

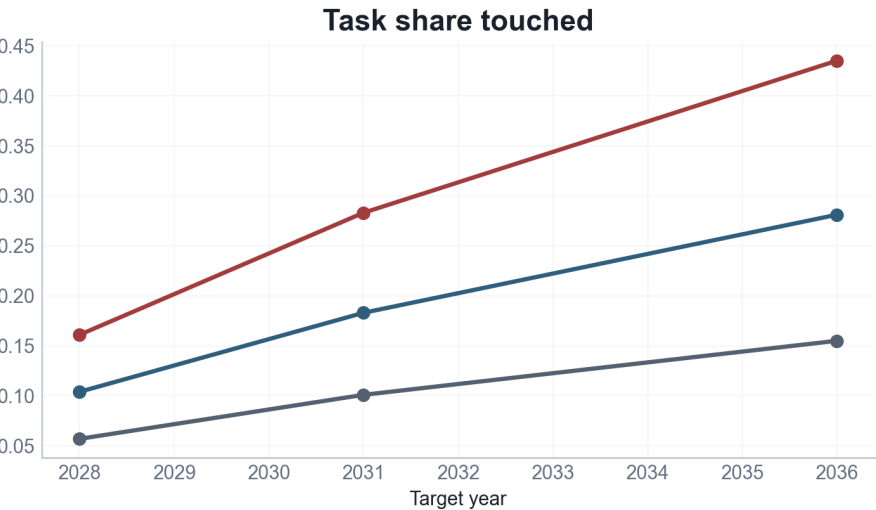
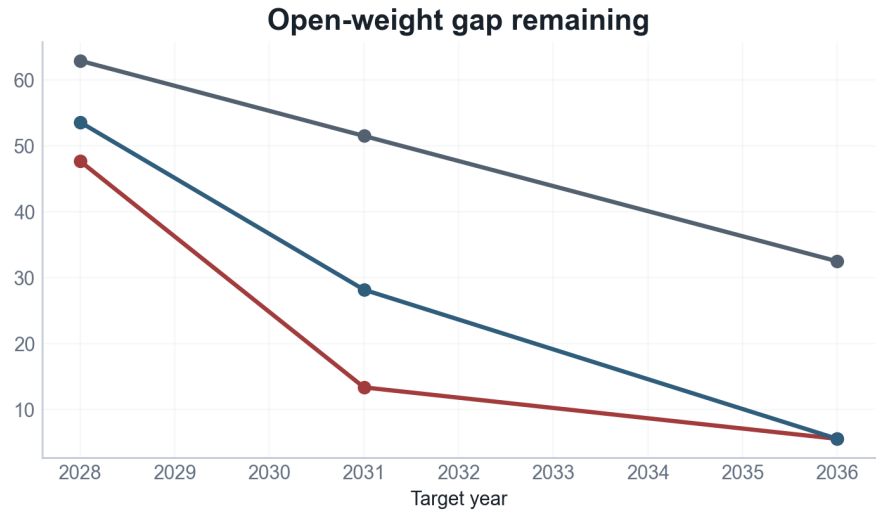
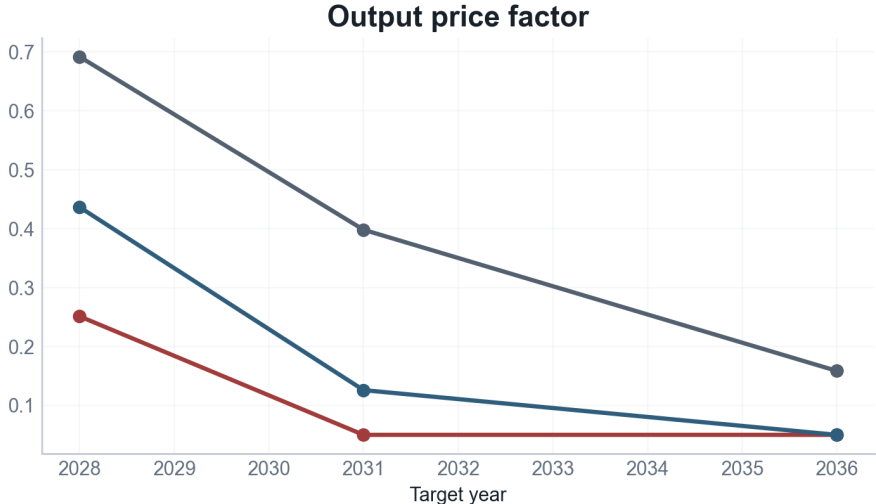
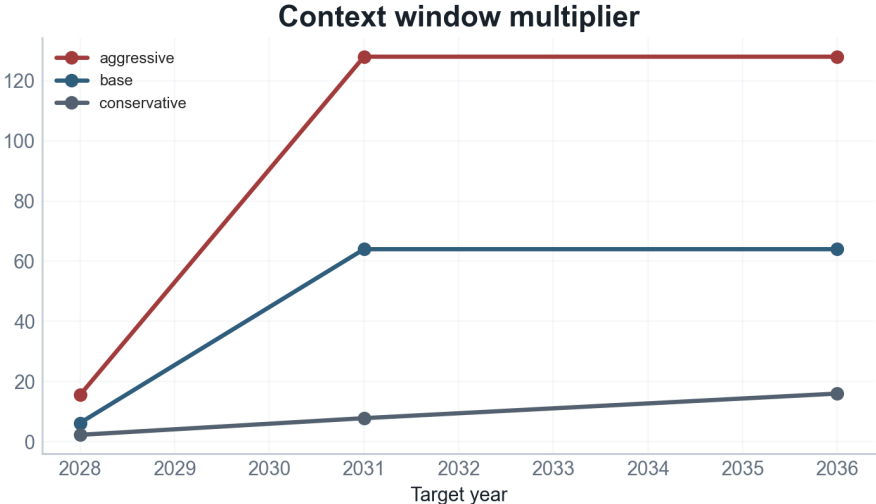
2, 5 And 10 Year Forecasts

Base scenario subset:

target_year	metric	value	unit	method
2028	frontier_context_window_multiplier	6.23	x current API catalog trend	capped scenario from OpenRouter upper-tail slope; raw=6.23x capped=False
2028	frontier_output_price_factor	0.4365	fraction of current low-price frontier API output cost	Explicit price-decline scenario; current-catalog release-cohort diagnostic=0.036, scenario_assumed=-0.180
2028	open_weight_lmarena_gap_remaining	53.6	arena rating points	current open vs closed LMArena gap with scenario-specific closure speed
2028	share_of_us_occupation_tasks_materially_touched	0.104	share of task-weighted occupation activity	Anthropic observed exposure plus O*NET task bottleneck pressure, scaled by horizon
2031	frontier_context_window_multiplier	64	x current API catalog trend	capped scenario from OpenRouter upper-tail slope; raw=96.70x capped=True
2031	frontier_output_price_factor	0.1259	fraction of current low-price frontier API output cost	Explicit price-decline scenario; current-catalog release-cohort diagnostic=0.036, scenario_assumed=-0.180
2031	open_weight_lmarena_gap_remaining	28.2	arena rating points	current open vs closed LMArena gap with scenario-specific closure speed
2031	share_of_us_occupation_tasks_materially_touched	0.183	share of task-weighted occupation activity	Anthropic observed exposure plus O*NET task bottleneck pressure, scaled by horizon
2036	frontier_context_window_multiplier	64	x current API catalog trend	capped scenario from OpenRouter upper-tail slope; raw=9351.33x capped=True
2036	frontier_output_price_factor	0.05	fraction of current low-price frontier API output cost	Explicit price-decline scenario; current-catalog release-cohort diagnostic=0.036, scenario_assumed=-0.180
2036	open_weight_lmarena_gap_remaining	5.6	arena rating points	current open vs closed LMArena gap with scenario-specific closure speed
2036	share_of_us_occupation_tasks_materially_touched	0.281	share of task-weighted occupation activity	Anthropic observed exposure plus O*NET task bottleneck pressure, scaled by horizon

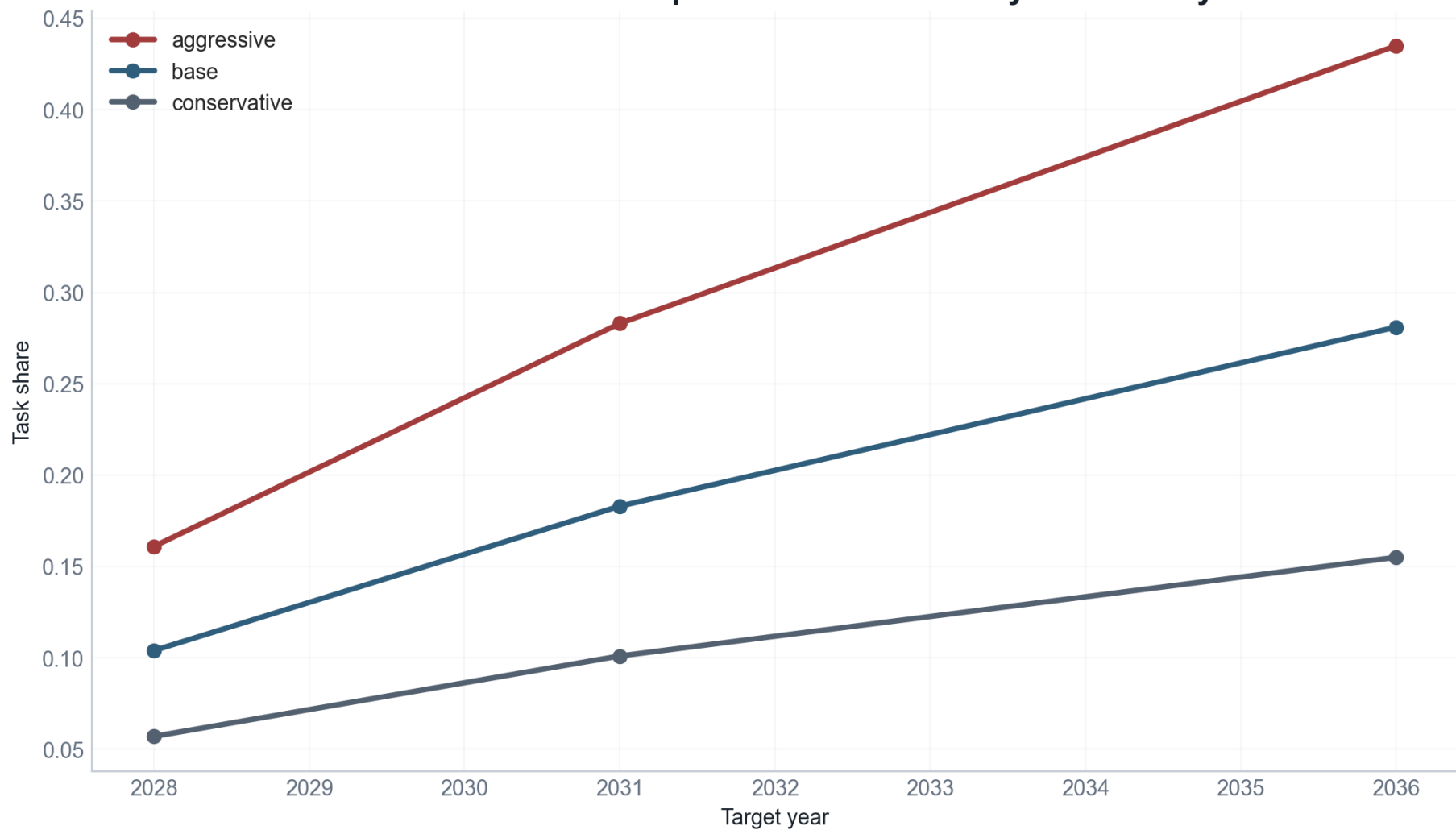
The dashboard puts four scenario families on one page: context scale, output price, open-weight benchmark gap and task-share contact. The useful reading is not the exact number in 2036; it is which assumptions move together and which do not.

Scenario Dashboard: 2, 5 and 10 Year Frontier-AI Pressure



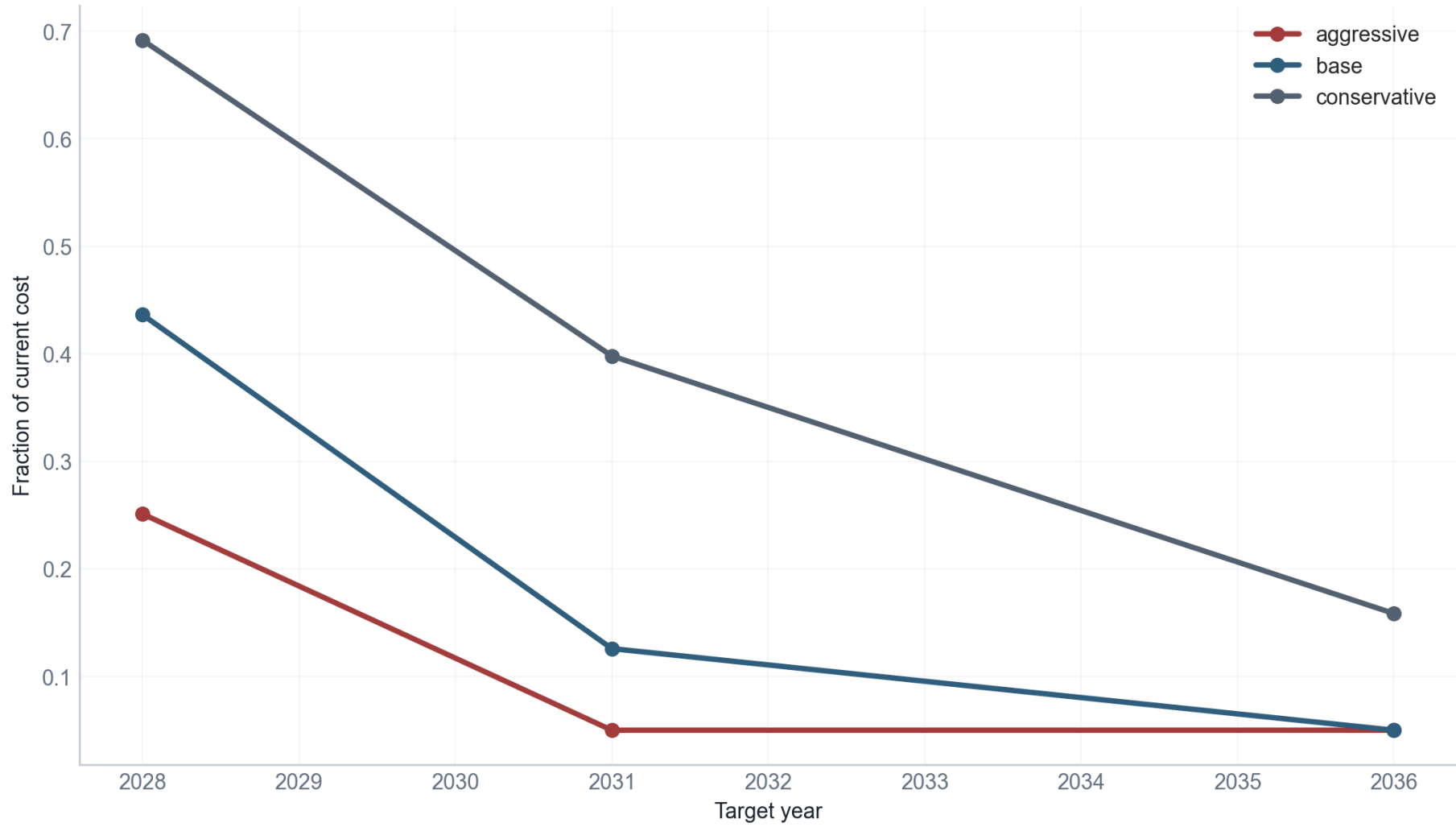
Forecast scenario dashboard

Scenario: Share of Occupation Tasks Materially Touched by AI



Labor task forecast

Scenario: Frontier Output Cost Factor



Cost forecast

Scenario: Open-Weight Gap Remaining



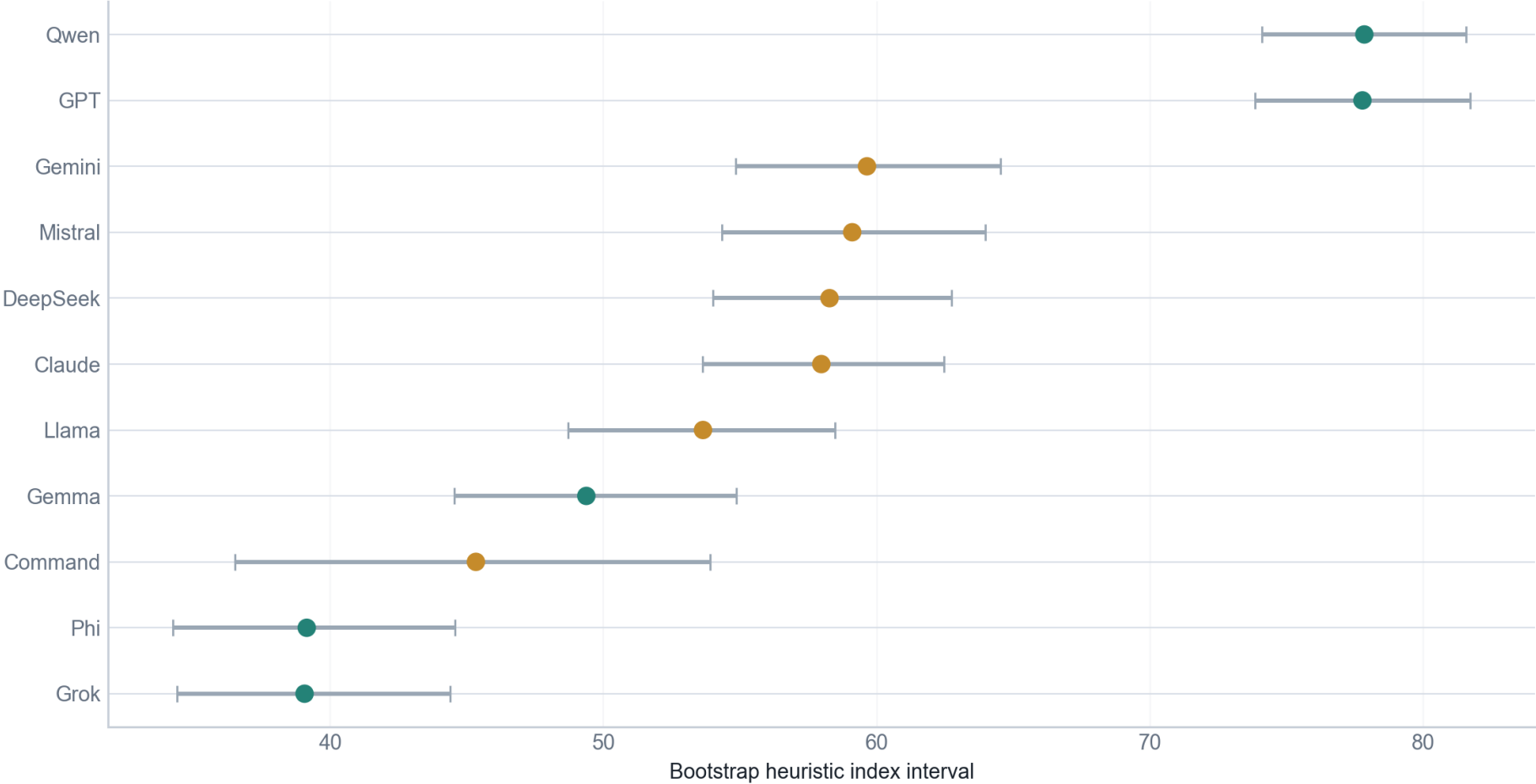
Open closed catchup

Uncertainty And Rank Stability

The uncertainty view stress-tests component weights and evidence depth. These intervals are not calibrated confidence intervals; they are a visibility layer for how much the rank can move when public-source signals are perturbed.

model_family	current_rank	score_p10	score_p50	score_p90	best_rank	median_rank	worst_rank	rank_stability_label
GPT	1	73.83	77.77	81.71	1	1	2	stable
Qwen	2	74.08	77.84	81.57	1	2	2	stable
Mistral	3	54.32	59.1	63.98	3	4	8	moderate
Gemini	4	54.83	59.64	64.52	3	4	9	moderate
Claude	6	53.63	57.97	62.46	3	5	8	moderate
DeepSeek	5	54	58.27	62.73	3	5	8	moderate
Llama	7	48.7	53.64	58.48	3	7	9	moderate
Gemma	8	44.54	49.37	54.86	3	8	10	stable
Command	9	36.51	45.33	53.9	3	9	11	moderate
Grok	11	34.4	39.06	44.38	7	10	11	stable
Phi	10	34.25	39.14	44.56	8	10	11	stable

Frontier Rank Uncertainty

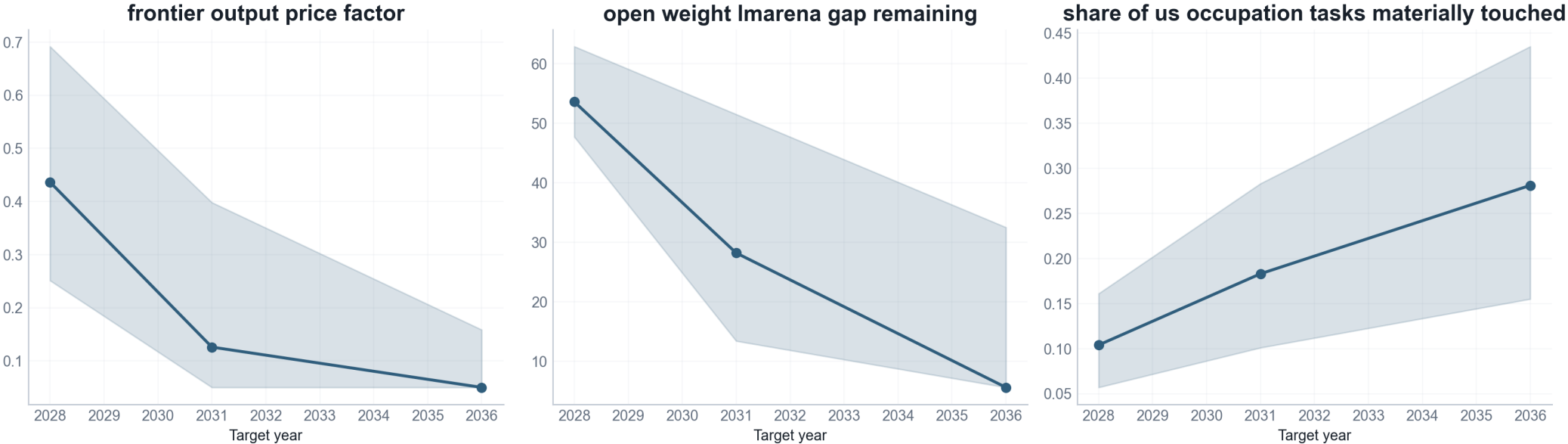


Intervals are evidence-scaled bootstrap stress tests, not calibrated confidence intervals. Labels come from rank IQR.

Frontier rank uncertainty

The forecast band chart is a scenario envelope across conservative, base and aggressive cases. It should not be read as a statistical confidence band.

Forecast Scenario Envelopes, Not Calibrated Confidence Bands



Forecast uncertainty bands

Release Velocity And Product Cadence

Release cadence separates visible public execution speed from benchmark quality. The cadence tables combine OpenRouter API catalog entries and Epoch metadata, deduplicated by family, vendor, product line and date.

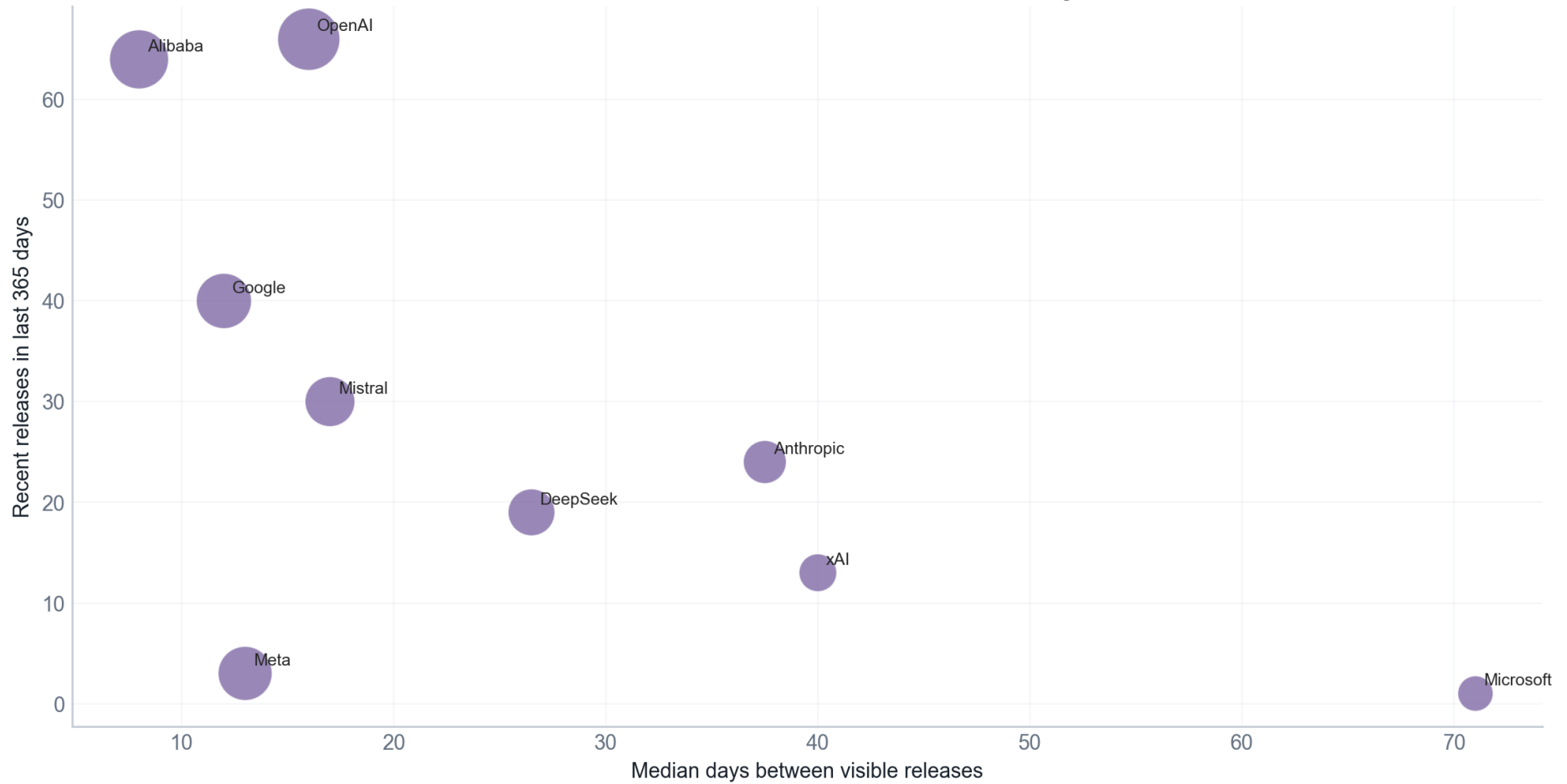
Family cadence:

model_family	vendor	total_releases	recent_releases_365d	median_days_between_releases	days_since_latest_release	cadence_label
GPT	OpenAI	162	66	16	10	fast
Qwen	Alibaba	131	64	8	18	fast
Gemini	Google	69	31	14	8	fast
Mistral	Mistral	67	30	17	15	fast
Claude	Anthropic	37	24	37.5	3	fast
DeepSeek	DeepSeek	52	19	26.5	21	fast
Grok	xAI	22	13	40	15	fast
Gemma	Google	31	9	18	42	fast
Llama	Meta	92	3	13	212	fast
Phi	Microsoft	17	1	71	210	fast

Vendor cadence:

vendor	portfolio_families	total_releases	recent_releases_365d	median_days_between_releases	cadence_label
OpenAI	GPT	162	66	16	fast
Alibaba	Qwen	131	64	8	fast
Google	Gemini,Gemma	100	40	12	fast
Mistral	Mistral	67	30	17	fast
Anthropic	Claude	37	24	37.5	fast
DeepSeek	DeepSeek	52	19	26.5	fast
xAI	Grok	22	13	40	fast
Meta	Llama	92	3	13	fast
Microsoft	Phi	17	1	71	fast

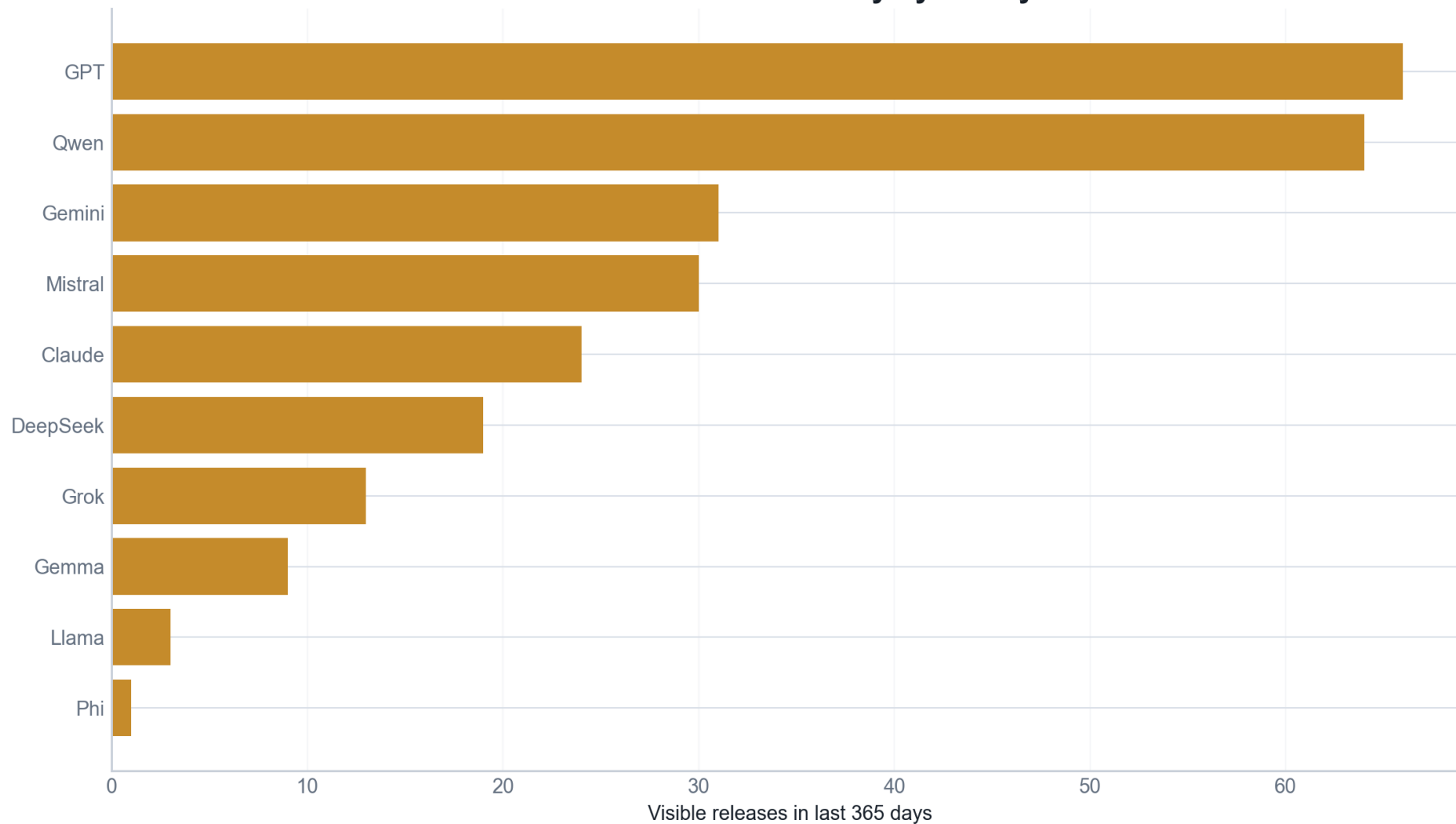
Release Cadence Timeline Summary



Bubble size tracks total visible releases. This is visible public cadence, not a complete internal product roadmap.

Release cadence timeline

Recent Release Velocity by Family



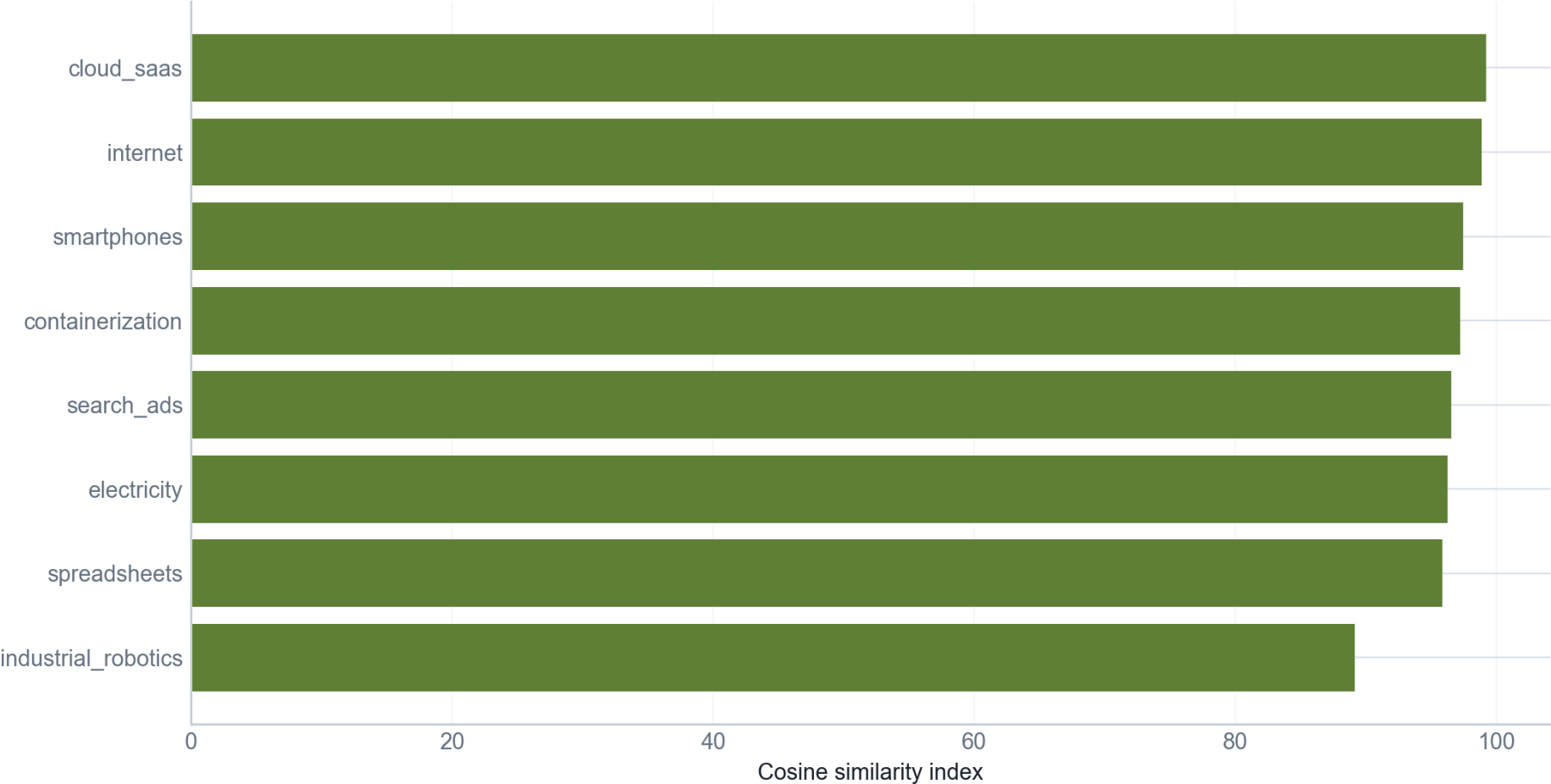
Recent release velocity

Historical Analogy

AI looks less like a single prior wave and more like an uncomfortable hybrid: spreadsheet-style task rebundling, internet-style diffusion, cloud-style API economics, and electricity-style long-run production redesign.

wave	period	ai_similarity_score	interpretation
cloud_saas	2006-2022	99.17	Best analogy for enterprise adoption lags and API-first business-model shift.
internet	1993-2010	98.86	Best analogy for general-purpose diffusion, platform creation and strange second-order labor demand.
smartphones	2007-2020	97.44	Best analogy for consumer pull, app ecosystems and fast behavioral rewiring.
containerization	1956-1990	97.23	Best analogy for cost shock in a hidden infrastructure layer.
search_ads	1998-2015	96.53	Best analogy for advertising-funded discovery and winner-take-most information layers.
electricity	1882-1930	96.22	Best analogy for long-run production reorganization, not near-term speed.
spreadsheets	1979-1995	95.85	Best analogy for occupational task rebundling and sudden knowledge-worker productivity jumps.
industrial_robotics	1961-2020	89.11	Useful negative analogy: physical deployment is slower and more capital-locked than software AI.

Historical Technology Wave Similarity to Frontier AI



This is a structured analogy index, not evidence of destiny. It frames where AI resembles prior technology waves and where it differs.

Historical analogy

Forecast Claims

claim_id	claim	evidence	confidence	analysis_captured_at
company-next-best-model	GPT has the strongest composite signal for near-term frontier leadership, but the top open-weight ecosystem score is not necessarily the same family.	Composite of LMArena, SWE-bench, OpenRouter, Epoch, Hugging Face, GitHub and OpenAlex indicators.	medium	2026-07-12T19:42:51+00:00
jobs-augmentation-not-total-replacement	The labor signal is broad task contact, not full-job deletion: high-exposure occupations still retain bottlenecks from trust, regulation, physical work and accountability.	Anthropic Economic Index occupation exposure joined to O*NET task text, task collaboration modes and wage/job metadata.	medium-high	2026-07-12T19:42:51+00:00
open-source-catchup	Open-weight systems look structurally advantaged on ecosystem and cost but still need repeated frontier jumps to erase closed/API benchmark gaps.	OpenRouter price fields, Hugging Face downloads/files, LMArena access-class split and Epoch open-weight release metadata.	medium	2026-07-12T19:42:51+00:00
ten-year-forecast	The 10-year question is less whether AI touches most cognitive workflows and more whether institutions redesign jobs around verification, liability and human preference.	Scenario table combines capability trend, price decline, observed task exposure and bottleneck scoring.	speculative	2026-07-12T19:42:51+00:00

Counterintuitive Findings

finding	evidence	why_it_is_interesting	artifact	analysis_captured_at
Raw frontier leadership and open-distribution upside are different questions.	In the 10-year frontier-quality scenario, GPT leads (58.6% of simulation draws); in the open-ecosystem-upside scenario, Qwen leads (70.2% of simulation draws).	The previous single 10-year number was misleading because it mixed best-model simulation share with adoption economics.	company_next_frontier_probabilities.csv	2026-07-12T19:42:51+00:00
The open-vs-closed gap is not one gap.	The largest measured LMArena category gap is text_to_image at 361.6 rating points.	Open-source catchup can be true in one domain and false in another; a single headline benchmark hides where closed labs still have moat.	open_closed_gap_by_category.csv	2026-07-12T19:42:51+00:00
Cheap models can sit on the efficient frontier without being the raw best model.	Efficient frontier examples include OpenAI: gpt-oss-20b; inclusionAI: Ling-2.6-flash.	Enterprise adoption often follows sufficient capability per dollar, not absolute leaderboard rank.	price_performance_frontier.csv	2026-07-12T19:42:51+00:00
The top whole-job automation candidates are narrower than the top task-exposure jobs.	The highest replacement-feasibility occupation is Market Research Analysts and Marketing Specialists with feasibility index 39.0.	A job can be heavily touched by AI but still mostly redesigned around human review rather than deleted.	job_replacement_feasibility.csv	2026-07-12T19:42:51+00:00
Augmentation can be a larger labor-weighted mode than replacement.	Available labor-weight proxy: augmentation-first=4,213,430, replacement-candidate=22,338.	This pushes the labor forecast toward workflow redesign, wage compression and productivity dispersion before mass full automation.	labor_market_exposure_summary.csv	2026-07-12T19:42:51+00:00

Where This Analysis Is Weak

The skeptical section names assumptions that could break the analysis. It is meant to make the work easier to challenge, not to protect it with broad caveats.

claim_id	assumption	failure_mode	mitigation	severity
family-frontier-ranking	Benchmark, release, ecosystem, cost and openness signals are directionally informative.	A private model or undisclosed benchmark result changes the frontier without appearing in public data.	Keep heuristic label, show sensitivity, and refresh source snapshots before external use.	high
direct-model-evidence	Normalized names and curated aliases correctly identify equivalent model rows.	Vendor naming drift or hidden routing makes same-looking model IDs non-equivalent.	Expose match confidence and keep unmatched/family-only rows in the audit table.	high
vendor-portfolio-ranking	Flagship plus evidence-weighted portfolio mean is a reasonable vendor aggregation.	A vendor has one dominant model family and several weak rows that should not affect portfolio interpretation.	Report flagship family, family count and components beside the vendor score.	medium
labor-domain-pressure	Keyword domain mapping is sufficient for a portfolio-level translation layer.	Occupation titles hide domain-specific workflows or regulated subdomains.	Keep occupation examples and human gates visible for each domain.	medium

claim_id	assumption	failure_mode	mitigation	severity
forecast-envelope	Conservative/base/aggressive scenarios bound the intended stress test.	A structural market break makes historical slopes irrelevant.	Label forecast bands as non-calibrated scenario envelopes.	high

Under-observed family audit:

model_family	vendor	direct_benchmark_match_count	coverage_score	underobserved	underobserved_reasons
Command	Command	0	0	True	few_direct_model_matches,low_source_coverage ,below_median_evidence_depth
GPT	OpenAI	120	99.5	True	below_median_evidence_depth
Gemini	Google	24	100	True	below_median_evidence_depth
Claude	Anthropic	48	100	True	below_median_evidence_depth
Grok	xAI	3	100	True	below_median_evidence_depth
Gemma	Google	15	96.3	False	none
Llama	Meta	81	98.08	False	none
DeepSeek	DeepSeek	25	98.89	False	none
Qwen	Alibaba	83	99.3	False	none
Mistral	Mistral	56	99.33	False	none
Phi	Microsoft	6	100	False	none

Method Notes

- Model-family scoring uses data/dataset/: LMArena full leaderboard rows, SWE-bench submissions, Open LLM Leaderboard metrics, OpenRouter prices/context, Epoch model metadata, Hugging Face rollups, GitHub model mentions and OpenAlex paper mentions.
- Domain scoring adds downloaded public benchmark sources under data/raw/domain_benchmarks/: LiveCodeBench, Open Medical-LLM, Terminal-Bench, FinanceBench, QFBench and Lexometrica. These are normalized into domain_benchmark_results.csv.
- Direct model evidence uses conservative name matching across exact, normalized exact, alias, family-only and unmatched classes. Family-only rows are audit evidence, not direct model proof.
- Vendor scoring maps model families to legal vendors and combines flagship-family signal with evidence-weighted portfolio breadth.
- Rank stability and forecast bands are stress tests and scenario envelopes. They are not calibrated confidence intervals.
- Labor scoring uses Anthropic Economic Index files from Hugging Face, including occupation exposure, task penetration, task automation/augmentation labels, O*NET task mappings/statements, and BLS wage/employment companion data.
- Scenario forecasts are not forecasts from a proprietary model. They are transparent transforms of observed slopes and pressure scores. Every scenario row includes a method field and the input diagnostics include caps/fallback policy.
- Domain forecasts use bounded gap closure from dated public benchmark frontier trends. When a domain lacks enough longitudinal evidence, the forecast uses a cross-domain fallback and marks confidence as low.
- Cost-per-message analysis is a workload-mix model over listed API price cohorts, not observed billing data. It separates low-cost chat, knowledge work, long-context analysis and agentic workflow runs.
- Fixed-task cost curves hold task token budgets and quality thresholds stable, then ask which current or future adequate family proxy is cheapest. They should be read as deployability screens, not direct model guarantees.

- Leadership simulation shares are stochastic sensitivity analyses over explicit score components, not calibrated market probabilities.
- Labor-weighted summaries use the best available public companion weights; where only major-group BLS employment is available, the analysis allocates it across detailed occupations inside that group to avoid treating each detailed occupation as the whole major group.
- BLS web xlsx endpoints returned anti-bot 403 responses in this environment. The analysis therefore uses public BLS-derived companion files already included in Anthropic's release rather than scraping around that restriction.

Generated Artifacts

- data/analysis/company_frontier_scores.csv
- data/analysis/dashboard_key_findings.csv
- data/analysis/domain_benchmark_catalog.csv
- data/analysis/domain_benchmark_results.csv
- data/analysis/domain_capability_frontier.csv
- data/analysis/domain_improvement_velocity.csv
- data/analysis/domain_capability_forecasts.csv
- data/analysis/domain_forecast_thresholds.csv
- data/analysis/company_score_methodology.csv
- data/analysis/company_score_sensitivity.csv
- data/analysis/model_benchmark_match_audit.csv
- data/analysis/direct_model_price_performance.csv
- data/analysis/llm_message_cost_trends.csv
- data/analysis/llm_message_cost_profile_components.csv
- data/analysis/fixed_task_cost_candidates.csv
- data/analysis/fixed_task_cost_curves.csv
- data/analysis/cost_divergence_scenarios.csv
- data/analysis/cost_external_evidence.csv
- data/analysis/vendor_frontier_scores.csv
- data/analysis/vendor_score_components.csv
- data/analysis/source_coverage_diagnostics.csv
- data/analysis/family_coverage_matrix.csv
- data/analysis/frontier_score_bootstrap.csv

- data/analysis/rank_stability_intervals.csv
- data/analysis/claim_failure_modes.csv
- data/analysis/underobserved_family_audit.csv
- data/analysis/business_domain_ai_pressure.csv
- data/analysis/domain_workflow_examples.csv
- data/analysis/release_cadence_by_family.csv
- data/analysis/release_cadence_by_vendor.csv
- data/analysis/job_exposure_scores.csv
- data/analysis/capability_forecasts.csv
- data/analysis/forecast_input_diagnostics.csv
- data/analysis/company_next_frontier_probabilities.csv
- data/analysis/open_closed_gap_by_category.csv
- data/analysis/lmarena_category_leaders.csv
- data/analysis/price_performance_frontier.csv
- data/analysis/labor_cluster_profiles.csv
- data/analysis/labor_market_exposure_summary.csv
- data/analysis/job_replacement_feasibility.csv
- data/analysis/counterintuitive_findings.csv
- data/analysis/historical_analogy_index.csv
- data/analysis/forecast_claims.csv
- figures/deep_analysis/company_score_component_stack.png
- figures/deep_analysis/company_score_evidence_scatter.png
- figures/deep_analysis/domain_benchmark_coverage.png
- figures/deep_analysis/domain_source_matrix.png
- figures/deep_analysis/domain_frontier_trends.png
- figures/deep_analysis/domain_current_velocity.png
- figures/deep_analysis/domain_forecast_base.png
- figures/deep_analysis/domain_forecast_scenarios.png
- figures/deep_analysis/domain_threshold_timeline.png
- figures/deep_analysis/leadership_scenario_matrix.png

- `figures/deep_analysis/open_closed_category_levels.png`
- `figures/deep_analysis/price_context_rating_map.png`
- `figures/deep_analysis/labor_outcome_mix.png`
- `figures/deep_analysis/forecast_scenario_dashboard.png`
- `figures/deep_analysis/direct_vs_proxy_price_performance.png`
- `figures/deep_analysis/llm_message_cost_trends.png`
- `figures/deep_analysis/fixed_task_cost_curves.png`
- `figures/deep_analysis/cost_task_message_divergence.png`
- `figures/deep_analysis/fixed_task_quality_cost_ladder.png`
- `figures/deep_analysis/vendor_frontier_scores.png`
- `figures/deep_analysis/family_vs_vendor_rank_shift.png`
- `figures/deep_analysis/source_coverage_dashboard.png`
- `figures/deep_analysis/family_signal_coverage_heatmap.png`
- `figures/deep_analysis/frontier_rank_uncertainty.png`
- `figures/deep_analysis/forecast_uncertainty_bands.png`
- `figures/deep_analysis/business_domain_pressure_matrix.png`
- `figures/deep_analysis/release_cadence_timeline.png`
- `figures/deep_analysis/recent_release_velocity.png`
- `figures/deep_analysis/*.png`